
Driving Post-Training Gains via Front-Loaded Reasoning Despite a Permissive-Data Constraint

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Abstract

1 Permissively licensed pre-training corpora make open foundation model research
2 lawful, auditable, and shareable, but they are widely assumed to cost capability.
3 Recent work narrowed that gap at the base-model stage by front-loading instruction
4 and reasoning data into pre-training, which leaves two variables bundled: the
5 licensing of the source pool and the composition of the mixture. We separate them
6 with a 3×2 factorial at 1.7B parameters and 300B pre-training tokens, crossing three
7 web substrates (a permissive-first pool, Nemotron-CC-HQ, and FineWeb-Edu) with
8 the presence or absence of the same instruction-and-reasoning subset, and carrying
9 every resulting model through an identical long-context extension and Tulu3 post-
10 training pipeline under one evaluation protocol with three decoding seeds. Front-
11 loading is the operative variable, raising the eleven-benchmark reasoning suite
12 average in all eighteen cells of the design by 10.3 to 19.2 points, with gains of
13 +23.0 points on code, +15.6 on mathematics, and +10.4 on instruction following.
14 Licensing is not. Once both pools are front-loaded the permissive substrate is
15 statistically indistinguishable from both non-permissive substrates at 4K context
16 after post-training and is ahead by 4.7 to 5.0 points at 16K, with the front-loading by
17 licensing interaction indistinguishable from zero. Receptivity to a later reasoning
18 stage is likewise a property of the mixture. Removing OpenThoughts3 from a 300B
19 pre-training run and then applying identical reasoning post-training drops the suite
20 average from 33.6 to 23.9 and returns AIME24 and AIME25 to zero, yet the gain
21 is not specific to that corpus and reappears with three held-out reasoning corpora.

22 1 Introduction

23 Modern language models are built in stages, and the boundary between them is moving. Instruction
24 data, mathematics, code, and reasoning traces, once the exclusive material of post-training, now
25 enter the pre-training mixture directly. This reopens a question that the staged view took for granted,
26 namely what a base model must already contain for post-training to do its work, and which later-stage
27 gains were latent in the pre-training corpus all along.

28 A second question runs alongside it. Open foundation model research has a practical interest in
29 permissively licensed corpora, which can be audited, redistributed, and built on without legal friction.
30 CommonPile [Kandpal et al., 2025] and CommonCorpus [Langlais et al., 2026] pursue exactly this,
31 historically at a measurable cost in capability, and MixtureVitae [Nguyen et al., 2025] narrowed that
32 cost substantially at the base-model stage by combining permissive-first sourcing with an unusually
33 large instruction and reasoning fraction. That result leaves two variables bundled. A permissive-first
34 mixture that is also reasoning-dominant may be competitive because of its licensing constraint,
35 because of its composition, or because of an interaction between them.

We separate them with a 3×2 factorial at 1.7B parameters and a 300B-token budget that crosses the *web substrate*, which carries the licensing constraint, with *front-loading*, the presence or absence of the same instruction-and-reasoning subset in the pre-training mixture. All six models are pre-trained under one fixed recipe, carried through an identical long-context extension and Tulu3 post-training pipeline, and evaluated under a single protocol with three decoding seeds and item-level bootstrap intervals. Because only two factors move, and the post-training recipe never does, the resulting effects are attributable to the pre-training mixture rather than to a co-varying stage boundary, hyperparameter, or curriculum shift.

The answer is asymmetric, and the asymmetry is the contribution. Front-loading is the driver, for permissive and non-permissive substrates alike, and licensing is a constraint under which that mechanism remains almost entirely available on reasoning and only partially available on general language understanding.

- **A factorial that separates composition from licensing.** At 300B pre-training tokens, under one recipe and one evaluation protocol with three seeds, front-loading raises the reasoning suite average in all eighteen cells of the design, by 10.3 to 19.2 points, with every bootstrap interval excluding zero (Section 3).
- **Licensing is nearly free on reasoning, and the interaction is null.** Once both pools are front-loaded, the permissive substrate matches both non-permissive substrates at 4K context after post-training, +0.5 and +0.4 points with intervals spanning zero, and leads at 16K by +5.0 and +4.7 points. The front-loading effect does not depend on licensing.
- **Licensing is not free on general language understanding.** Front-loading helps every substrate but does not close the substrate gap, which remains at 6.8 to 8.8 points.
- **Receptivity is a property of the mixture, not of the post-training corpus.** Removing OpenThoughts3 from a 300B pre-training run and then applying identical reasoning post-training costs 9.7 points of suite average and 31.8 points of MATH500, yet the gain reappears with three reasoning corpora held out of pre-training (Section 5).
- **Open artifacts.** All base, SFT, DPO, and reasoning post-trained checkpoints will be released, with the pipeline that regenerates every table and figure here from per-sample evaluation records.

Front-loading is an established effect, shown by Front-Loaded Reasoning [Akter et al., 2025], by mid-training work such as PRISM [Runwal et al., 2026] and Zhang et al. [2026], and anticipated by LIMA [Zhou et al., 2023a] and LIMO [Ye et al., 2025], where very small post-training sets elicit capability the base must already have held. What is new here is measuring it in a design where the attribution is unambiguous, in single-stage pre-training with full decontamination, and under a licensing constraint that no prior study has varied.

2 Experimental setup

2.1 Factorial design and matched pre-training

Every model in the factorial is a 1.7B decoder transformer trained on 300B tokens under the *open-sci-ref* protocol [Nezhurina et al., 2026], which fixes architecture, tokenizer, optimizer, learning rate schedule, batch size, and total tokens, so that models differ only in their pre-training data. Full hyperparameters are in Appendix C.1.

The two factors are the following.

Web substrate takes three levels. The permissive-first substrate is the curated web portion of MixtureVitae [Nguyen et al., 2025]. The two non-permissive substrates are Nemotron-CC-HQ [Su et al., 2025], the strongest pre-training dataset identified in the *open-sci-ref* evaluation, and FineWeb-Edu [Penedo et al., 2024]. The licensing constraint attaches to the substrate, so a model trained on a non-permissive substrate is non-permissive overall even where the front-loaded subset itself is permissive.

Front-loading takes two levels, namely whether the MixtureVitae instruction-and-reasoning subset is present in the pre-training mixture. This is the same subset in all three substrates, which is what makes the front-loading effect comparable across them.

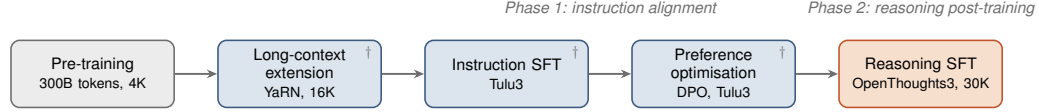


Figure 1: **Post-training pipeline.** Stages marked † are skipped in some experiments and for some models. Bases with a native context of 16K or more skip the extension stage, the factorial and mixture-ablation models stop after DPO, and reasoning post-training is applied from the SFT checkpoint by default and from the base and DPO checkpoints in the controls of Section 5.

Crossing the two factors gives six pre-training runs, each produced in a 4K and a 16K context variant and evaluated at three post-training stages, for thirty-six evaluated checkpoints and eighteen cells of the front-loading design. The pool is roughly 422B tokens, which permits a 300B-token budget without repetition.

2.2 Model zoo

Beyond the factorial we post-train a wider set of 1.7B bases under the same pipeline. The matched-compute references are trained on 300B tokens under the identical `open-sci-ref` recipe, namely Nemotron-CC-HQ, FineWeb-Edu, DCLM [Li et al., 2024a], and CommonPile v0.1 [Kandpal et al., 2025] (denoted Comma0.1, permissively licensed but without the instruction and reasoning enrichment). The external references are SmolLM2 1.7B [Allal et al., 2025], Qwen2.5 1.5B [Qwen et al., 2025], and Qwen3 1.7B [Yang et al., 2025], released under their own recipes at 11T, 18T, and 36T tokens, and are uncontrolled upper references rather than matched comparisons, since architecture, tokenizer, and pre-training compute all differ.

2.3 Post-training pipeline

Figure 1 gives the pipeline. Post-training data is identical across models within each stage.

Long-context extension. We extend the context window from 4K to 16K with YaRN [Peng et al., 2024], using a scaling factor of 4.0 and one epoch of full-parameter supervised fine-tuning at a learning rate of 2×10^{-4} with cosine decay. The mixture is 30% long-context data, combining SeaLong [Li et al., 2024b] and LongAlign [Bai et al., 2024], and 70% SmolTalk [Allal et al., 2025], which preserves general instruction following. The same recipe is applied to every base, so long-context behavior is attributable to the base rather than to the recipe.

Phase 1, instruction alignment. We apply the Tulu3 [Lambert et al., 2025] recipe in its two standard stages, supervised fine-tuning on the Tulu3 SFT mixture followed by DPO [Rafailov et al., 2023] on Tulu3 preference data, both at Tulu3 defaults without model-specific tuning. We do not run Tulu3’s reinforcement learning with verifiable rewards stage, so all Phase 1 results here describe the SFT and DPO stages only.

Phase 2, reasoning post-training. We train for five epochs on a uniformly sampled 30K subset of OpenThoughts3 [Guha et al., 2025] with the OT3 medium hyperparameters. The default starting point is each model’s Phase 1 SFT checkpoint, since SFT and DPO give similar reasoning scores. The controls in Section 5 additionally start from the post-YaRN base and from the DPO checkpoint.

2.4 Evaluation

Suites. The primary suite covers mathematics with AIME24 [MAA, 2024], AIME25 [MAA, 2025], AMC23 [MAA, 2023], MATH500 [Hendrycks et al., 2021], and GSM8K [Cobbe et al., 2021], code with HumanEval [Chen et al., 2021], LiveCodeBench [Jain et al., 2025], and MBPP [Austin et al., 2021], science and knowledge-intensive question answering with GPQA-Diamond [Rein et al., 2024] and JEEBench [Arora et al., 2023], and instruction following with IFEval [Zhou et al., 2023b]. The suite average is an unweighted macro average over these eleven tasks, and we treat it as a diagnostic to be read alongside the per-benchmark and per-domain views rather than as a model-quality metric. As a secondary safeguard we evaluate eleven standard language-understanding tasks with the LM Evaluation Harness [Gao et al., 2021] following Nguyen et al. [2025], listed with shot counts in Table 28. Those runs are single seed, so the language results carry no error bars.

Table 1: **Front-loading versus source licensing at 300B pre-training tokens.** Each *no* and *yes* row is one 1.7B model trained under the identical `open-sci-ref` configuration, differing only in the web substrate and in whether the MixtureVitae instruction-and-reasoning subset is included. Reasoning columns are the eleven-benchmark suite average under Protocol A (Appendix Table 26). Values are mean $\pm$ SD over decoding seeds 42, 1234, and 2025. Language columns are the eleven-task understanding suite on the corresponding base checkpoint, single seed. Δ rows give the front-loading effect in percentage points.

Web substrate	Front-loaded	Reasoning, 4K			Reasoning, 16K			Language	
		Base	SFT	DPO	Base	SFT	DPO	4K	16K
MixtureVitae (<i>permissive</i>)	no	7.4 $\pm$ 0.02	11.1 $\pm$ 0.03	11.5 $\pm$ 0.09	11.6 $\pm$ 0.03	11.8 $\pm$ 0.02	12.0 $\pm$ 0.15	51.0	50.8
	yes	17.7 $\pm$ 0.29	23.0 $\pm$ 0.16	23.9 $\pm$ 0.17	30.7 $\pm$ 0.72	27.3 $\pm$ 0.11	27.8 $\pm$ 0.04	56.0	55.8
	Δ	+10.3	+11.9	+12.4	+19.1	+15.5	+15.8	+5.0	+5.0
Nemotron-CC-HQ	no	3.9 $\pm$ 0.01	8.0 $\pm$ 0.04	8.3 $\pm$ 0.07	8.1 $\pm$ 0.08	8.4 $\pm$ 0.12	8.6 $\pm$ 0.06	64.2	64.1
	yes	23.1 $\pm$ 0.33	22.5 $\pm$ 0.10	23.5 $\pm$ 0.21	27.3 $\pm$ 0.38	22.3 $\pm$ 0.27	22.7 $\pm$ 0.27	64.6	64.6
	Δ	+19.2	+14.5	+15.2	+19.2	+13.9	+14.1	+0.4	+0.5
FineWeb-Edu	no	6.0 $\pm$ 0.06	7.1 $\pm$ 0.07	7.1 $\pm$ 0.07	8.1 $\pm$ 0.09	7.9 $\pm$ 0.07	8.5 $\pm$ 0.09	56.7	57.4
	yes	25.2 $\pm$ 0.07	22.6 $\pm$ 0.15	23.0 $\pm$ 0.08	27.1 $\pm$ 0.06	22.6 $\pm$ 0.24	22.1 $\pm$ 0.07	62.8	62.6
	Δ	+19.2	+15.5	+15.9	+19.0	+14.7	+13.6	+6.1	+5.2

Two evaluation protocols. Protocol A decodes greedily on all eleven benchmarks under one fixed generation budget, which is what makes thirteen model families and three decoding seeds comparable, and it covers Phase 1, the factorial, and the mixture ablations. Protocol B uses sampled decoding and longer generation budgets on several benchmarks, because models that have received reasoning post-training emit long chains of thought that a short greedy budget truncates, and it covers every OpenThoughts3 experiment. Full settings are in Tables 26 and 27. The two protocols score the same checkpoint differently, so no number from one is ever compared against a number from the other, and every Protocol B result is a paired change measured against a starting checkpoint re-evaluated under Protocol B. Protocol A results are the mean and sample standard deviation over decoding seeds 42, 1234, and 2025, whose residual variation under greedy decoding comes from batching and kernel nondeterminism in the inference engine. These are decoding seeds and do not estimate variability across independent pre-training runs. Protocol B results are single seed and carry no error bars.

Decontamination and statistics. The MixtureVitae pre-training data and all shared post-training data are decontaminated against every benchmark reported here with the 13-gram exact-match protocol of Nguyen et al. [2025], applied separately to the Tulu3 SFT mixture, the Tulu3 preference data, and the OpenThoughts3 subset. The non-permissive substrates and the external references are used as released and inherit whatever contamination they carry, an asymmetry that favors them and that Appendix E quantifies. Contrasts on the suite average carry 95% paired item-level bootstrap intervals from 2,000 replicates that resample items within each benchmark after averaging over seeds, with Benjamini-Hochberg correction within each contrast family. Because JEEBench awards partial credit, items are treated as scores in $[0, 1]$ rather than as Bernoulli outcomes. Appendix A gives the full contrast table.

3 Front-loading versus source licensing

Front-loading is the operative variable. Table 1 and Figure 2 give the design. Adding the instruction-and-reasoning subset to the pre-training mixture raises the reasoning suite average in all eighteen cells, by 10.3 to 19.2 points, with every bootstrap interval excluding zero at $q < 0.001$. At 16K after Tulu3 SFT the effect is +15.5 points for the permissive substrate ($11.8 \rightarrow 27.3$), +13.9 for Nemotron-CC-HQ ($8.4 \rightarrow 22.3$), and +14.7 for FineWeb-Edu ($7.9 \rightarrow 22.6$). It is largest at the base stage, near 19 points for every substrate at 16K, and Phase 1 recovers only part of the deficit its absence creates. Tulu3 contains instruction, mathematics, and code data of its own, and applying it to a base without front-loaded reasoning does not substitute for having pre-trained on it.

Licensing is nearly free on reasoning. Once both pools are front-loaded the substrate contrast largely disappears. At 4K after Tulu3 SFT the permissive substrate leads Nemotron-CC-HQ by +0.5 points ($[-1.2, +2.3]$, $q = 0.65$) and FineWeb-Edu by +0.4 points ($[-1.3, +1.9]$, $q = 0.66$), both indistinguishable from zero, and at 16K it leads by +5.0 ($[+3.2, +7.0]$) and +4.7 points

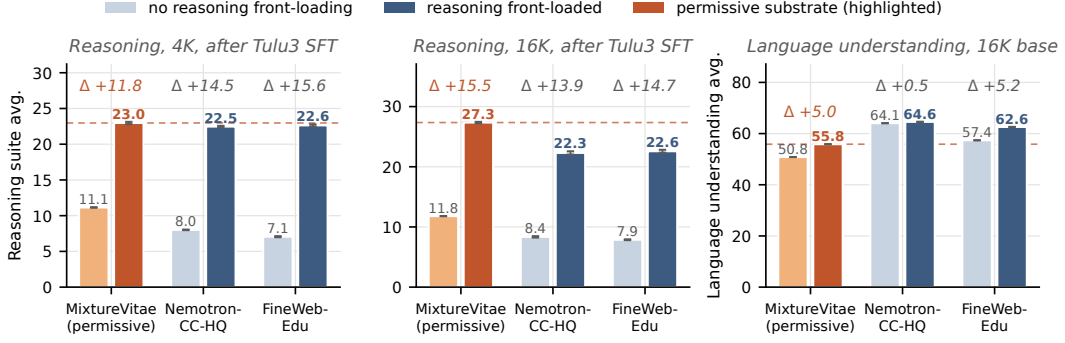


Figure 2: **Front-loading drives the reasoning gain and licensing does not.** Paired bars give the suite average without and with the instruction-and-reasoning subset in pre-training, for each web substrate, after Tulu3 SFT. The permissive substrate is highlighted, and the dashed line marks its front-loaded level so that the substrate contrast at matched front-loading can be read directly. Error bars are the sample standard deviation over three decoding seeds. Language understanding is single seed and measured on the base checkpoints.

($[+3.0, +6.6]$), both with $q \leq 0.001$. The only stage at which a non-permissive substrate is ahead is the 4K base, before any post-training, where it leads by 5.4 to 7.5 points. Post-training removes that advantage rather than preserving it.

The mechanism does not depend on licensing. The interaction between the two factors, the front-loading effect on the permissive substrate minus its mean effect on the two non-permissive substrates, is $+0.0$ points at the 16K base ($[-2.0, +2.0]$, $q = 0.97$), $+1.2$ after SFT ($[-0.8, +3.2]$, $q = 0.28$), and $+1.9$ after DPO ($[-0.1, +4.0]$, $q = 0.10$), none distinguishable from zero. The effect is the same size wherever it is applied, which is what supports reading front-loading as the mechanism and licensing as a separate constraint.

The gains are broad. Averaged over the three substrates at 16K after SFT, front-loading is worth $+23.0$ points on code, $+15.6$ on mathematics, $+10.4$ on instruction following, and $+2.5$ on science and knowledge-intensive question answering (Figure 6), and instruction following improves in all twelve post-training cells, by 7.7 to 17.5 points. Because an unweighted average over five mathematics and three code benchmarks implicitly weights those domains at 45% and 27%, we recompute both headline quantities under an equal-domain average and with each domain removed in turn, and neither changes materially (Table 6).

Language understanding is where the constraint is not free. Front-loading also helps general language understanding, by $+5.0$ points for the permissive substrate ($50.8 \rightarrow 55.8$ at 16K) and $+5.2$ for FineWeb-Edu ($57.4 \rightarrow 62.6$), though only $+0.5$ for Nemotron-CC-HQ, which already contains synthetically rephrased question-and-answer text and starts much higher. The instructive detail is what the non-front-loaded models cannot do. Without the subset, FineWeb-Edu and the permissive substrate score 26.1 and 25.8 on 5-shot MMLU and 21.1 and 22.4 on 10-shot CommonsenseQA, which is chance on both, and front-loading lifts them to 47.5 and 38.7 on MMLU and 55.7 and 48.7 on CommonsenseQA. Whatever else instruction and reasoning data supplies, it supplies the ability to answer a benchmark in the format the benchmark asks for.

It does not, however, close the substrate gap. At matched front-loading the permissive substrate reaches 55.8 against 64.6 for Nemotron-CC-HQ and 62.6 for FineWeb-Edu, a residual 6.8 to 8.8 points that front-loading does not recover in our design. Permissiveness is close to free on the capabilities post-training targets and is not free on broad language coverage.

4 Post-training across the model zoo

The factorial result survives the wider comparison. Figure 3 and Table 2 place the front-loaded permissive model against the bases the community actually trains on. After Phase 1 at 16K it

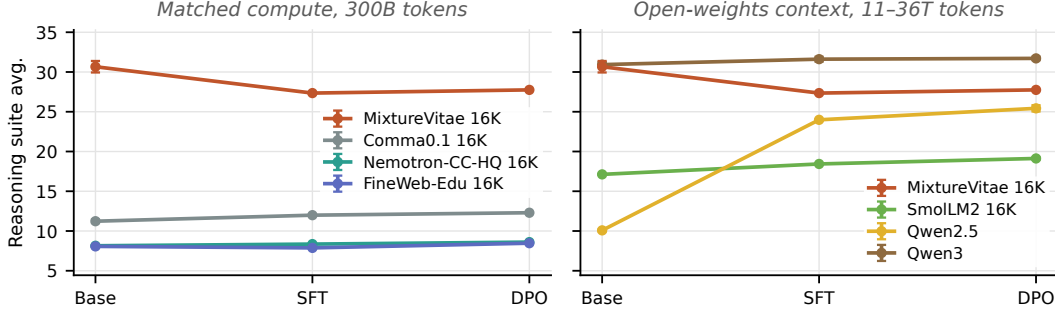


Figure 3: **Phase 1 trajectories, Protocol A.** Left, matched-compute bases trained on 300B tokens under the open-sci-ref protocol. Right, the same permissive model against open-weights references trained on 36 to 120 times more tokens, shown as uncontrolled upper references. Error bars are the sample standard deviation over three decoding seeds. Per-benchmark values are in Table 2.

Table 2: **Phase 1 across the model zoo.** One context variant per model family, namely the 16K variant where one exists and the released context otherwise. The matched-compute models are trained on 300B tokens under the open-sci-ref protocol, while SmolLM2, Qwen2.5, and Qwen3 use 11T, 18T, and 36T tokens respectively and are uncontrolled upper references. Both context variants of every model are in Table 7. Protocol A (Appendix Table 26). Values are mean $\pm$ SD over decoding seeds 42, 1234, and 2025.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
<i>Matched compute, 300B tokens, controlled comparison</i>												
MixtureVitae-16K	1.1 $\pm$ 1.92	3.3 $\pm$ 0.00	23.3 $\pm$ 8.04	47.0 $\pm$ 0.53	60.0 $\pm$ 0.00	54.1 $\pm$ 0.35	22.3 $\pm$ 0.34	40.1 $\pm$ 0.23	23.4 $\pm$ 0.29	8.0 $\pm$ 0.24	54.6 $\pm$ 0.23	30.7 $\pm$ 0.72
→ SFT	3.3 $\pm$ 0.00	3.3 $\pm$ 0.00	15.0 $\pm$ 0.00	35.0 $\pm$ 0.53	51.6 $\pm$ 0.00	54.1 $\pm$ 0.35	14.4 $\pm$ 0.11	40.4 $\pm$ 0.35	23.4 $\pm$ 0.29	5.1 $\pm$ 0.27	55.1 $\pm$ 0.26	27.3 $\pm$ 0.11
→ SFT → DPO	3.3 $\pm$ 0.00	1.1 $\pm$ 1.92	17.5 $\pm$ 0.00	37.0 $\pm$ 0.80	52.6 $\pm$ 0.00	52.6 $\pm$ 0.70	14.0 $\pm$ 0.30	40.3 $\pm$ 0.12	23.4 $\pm$ 0.29	6.2 $\pm$ 0.20	56.5 $\pm$ 0.74	27.8 $\pm$ 0.04
Nemotron-CC-HQ-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.8 $\pm$ 0.35	8.2 $\pm$ 0.00	1.8 $\pm$ 0.00	0.1 $\pm$ 0.11	5.7 $\pm$ 0.31	22.7 $\pm$ 0.51	5.6 $\pm$ 1.04	42.7 $\pm$ 0.38	8.1 $\pm$ 0.08
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.9 $\pm$ 0.31	8.6 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.9 $\pm$ 0.12	22.2 $\pm$ 0.87	1.2 $\pm$ 0.08	45.8 $\pm$ 0.00	8.4 $\pm$ 0.12
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.5 $\pm$ 0.12	7.7 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.3 $\pm$ 0.12	23.4 $\pm$ 1.17	1.7 $\pm$ 0.37	46.3 $\pm$ 0.61	8.6 $\pm$ 0.06
FineWeb-Edu-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.9 $\pm$ 0.12	8.6 $\pm$ 0.00	2.4 $\pm$ 0.00	0.1 $\pm$ 0.11	2.0 $\pm$ 0.00	24.9 $\pm$ 0.29	6.5 $\pm$ 0.58	38.8 $\pm$ 0.53	8.1 $\pm$ 0.09
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.1 $\pm$ 0.12	8.7 $\pm$ 0.00	4.9 $\pm$ 0.00	0.2 $\pm$ 0.00	2.5 $\pm$ 0.23	23.6 $\pm$ 0.58	2.4 $\pm$ 0.40	41.3 $\pm$ 0.46	7.9 $\pm$ 0.07
→ SFT → DPO	0.0 $\pm$ 0.00	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	3.3 $\pm$ 0.12	9.2 $\pm$ 0.00	4.9 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.12	23.9 $\pm$ 0.29	2.7 $\pm$ 0.54	43.3 $\pm$ 0.57	8.5 $\pm$ 0.09
DCLM	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.00	3.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	19.5 $\pm$ 0.77	0.0 $\pm$ 0.00	39.2 $\pm$ 0.08	5.6 $\pm$ 0.08
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.8 $\pm$ 0.00	5.2 $\pm$ 0.00	6.7 $\pm$ 0.00	0.2 $\pm$ 0.00	5.4 $\pm$ 0.35	21.7 $\pm$ 0.00	1.1 $\pm$ 0.49	40.6 $\pm$ 0.13	7.4 $\pm$ 0.02
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	1.1 $\pm$ 0.23	4.6 $\pm$ 0.00	6.7 $\pm$ 0.00	0.3 $\pm$ 0.11	5.6 $\pm$ 0.00	20.7 $\pm$ 0.00	0.5 $\pm$ 0.03	43.0 $\pm$ 0.08	7.7 $\pm$ 0.15
Comma0.1-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	2.8 $\pm$ 0.35	12.2 $\pm$ 0.00	13.4 $\pm$ 0.00	0.1 $\pm$ 0.11	16.0 $\pm$ 0.20	25.8 $\pm$ 1.01	7.8 $\pm$ 0.19	43.8 $\pm$ 0.81	11.2 $\pm$ 0.22
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.5 $\pm$ 0.12	6.9 $\pm$ 0.00	16.5 $\pm$ 0.00	1.4 $\pm$ 0.20	25.2 $\pm$ 0.35	23.2 $\pm$ 0.00	4.8 $\pm$ 0.66	49.0 $\pm$ 0.27	12.0 $\pm$ 0.01
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.1 $\pm$ 0.12	6.7 $\pm$ 0.00	17.1 $\pm$ 0.00	1.8 $\pm$ 0.23	25.9 $\pm$ 0.31	23.4 $\pm$ 0.29	5.1 $\pm$ 0.12	50.6 $\pm$ 0.23	12.3 $\pm$ 0.03
<i>External open-weights references, 11-36T tokens, uncontrolled</i>												
SmolLM2-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.8 $\pm$ 1.44	11.8 $\pm$ 0.92	31.9 $\pm$ 0.00	24.0 $\pm$ 0.35	1.2 $\pm$ 0.23	36.5 $\pm$ 0.12	24.6 $\pm$ 0.29	7.6 $\pm$ 0.39	49.9 $\pm$ 0.27	17.1 $\pm$ 0.07
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	9.6 $\pm$ 0.20	32.8 $\pm$ 0.00	32.9 $\pm$ 0.61	4.4 $\pm$ 0.11	37.5 $\pm$ 0.23	24.9 $\pm$ 1.91	4.7 $\pm$ 0.72	54.4 $\pm$ 0.40	18.4 $\pm$ 0.05
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.3 $\pm$ 1.44	9.6 $\pm$ 0.53	33.8 $\pm$ 0.00	32.3 $\pm$ 1.06	5.3 $\pm$ 0.20	36.9 $\pm$ 0.50	26.3 $\pm$ 0.00	5.4 $\pm$ 0.69	57.4 $\pm$ 0.20	19.1 $\pm$ 0.17
Qwen2.5	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.35	61.9 $\pm$ 0.00	57.0 $\pm$ 0.70	0.2 $\pm$ 0.00	0.1 $\pm$ 0.12	11.1 $\pm$ 1.82	0.3 $\pm$ 0.10	31.3 $\pm$ 0.34	10.1 $\pm$ 0.15
→ SFT	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	8.3 $\pm$ 1.44	18.8 $\pm$ 0.69	39.0 $\pm$ 0.00	51.0 $\pm$ 0.35	8.0 $\pm$ 0.23	44.3 $\pm$ 0.23	29.0 $\pm$ 0.77	6.1 $\pm$ 0.54	56.0 $\pm$ 0.27	24.0 $\pm$ 0.07
→ SFT → DPO	2.2 $\pm$ 1.92	0.0 $\pm$ 0.00	10.0 $\pm$ 2.50	21.2 $\pm$ 1.00	40.4 $\pm$ 0.00	55.7 $\pm$ 0.70	8.5 $\pm$ 0.23	44.5 $\pm$ 0.23	27.9 $\pm$ 0.58	8.0 $\pm$ 0.64	61.2 $\pm$ 0.30	25.4 $\pm$ 0.40
Qwen3	3.3 $\pm$ 0.00	2.2 $\pm$ 1.92	20.8 $\pm$ 3.82	48.0 $\pm$ 0.35	71.6 $\pm$ 0.00	64.2 $\pm$ 0.93	1.6 $\pm$ 0.11	46.5 $\pm$ 0.81	25.1 $\pm$ 0.77	18.0 $\pm$ 0.76	38.8 $\pm$ 0.34	30.9 $\pm$ 0.31
→ SFT	0.0 $\pm$ 0.00	16.7 $\pm$ 2.89	28.2 $\pm$ 0.35	69.4 $\pm$ 0.00	67.1 $\pm$ 0.61	67.1 $\pm$ 0.61	3.5 $\pm$ 0.85	54.3 $\pm$ 0.58	33.0 $\pm$ 1.05	11.9 $\pm$ 0.54	63.7 $\pm$ 0.13	31.6 $\pm$ 0.30
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	8.3 $\pm$ 1.44	31.2 $\pm$ 1.06	69.8 $\pm$ 0.00	67.7 $\pm$ 0.61	4.9 $\pm$ 0.00	54.7 $\pm$ 0.23	31.3 $\pm$ 0.87	13.3 $\pm$ 1.25	67.6 $\pm$ 1.33	31.7 $\pm$ 0.21

reaches 27.8, against 12.3 for Comma0.1, 8.6 for Nemotron-CC-HQ, and 8.5 for FineWeb-Edu at the same 300B budget, and 7.7 for DCLM at 4K. Comma0.1 is the informative comparison, since it is permissively licensed but not reasoning enriched, and it sits with the non-permissive references rather than with the front-loaded model. Licensing does not predict the ranking and composition does. Against the external references it exceeds SmolLM2 at 16K (19.1) and Qwen2.5 (25.4) and trails Qwen3 (31.7), which uses 120 times the tokens and whose pre-training already includes reasoning traces.

Short-context alignment erodes long-context reasoning. The permissive 16K model is strongest before Phase 1, at 30.7, and Tulu3 SFT lowers it to 27.3, with DPO recovering to 27.8 but not to the base level. This is a degradation of an unusually strong post-extension base rather than a weak post-trained model, since the result still leads every matched-compute reference by more than 15 points. The natural reading is that the short-context-dominated Tulu3 mixture partially overwrites reasoning behavior that long-context extension and reasoning-rich pre-training installed, and Section 5 tests that reading directly. The same pattern is visible for Qwen3, whose base leads its own post-trained checkpoints on MATH500, so it is not specific to one mixture.

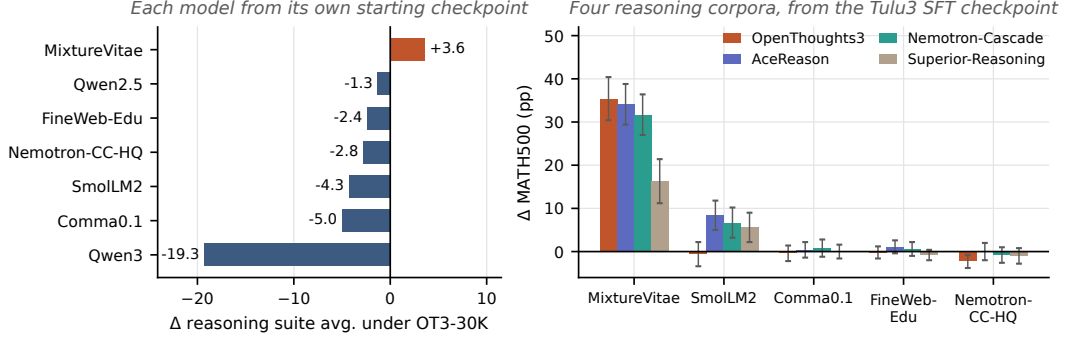


Figure 4: **Only the front-loaded permissive base converts a small reasoning budget into gains.** Left, change in suite average under identical OpenThoughts3-30K post-training, measured for each model from its own Tulu3 SFT checkpoint, except MixtureVitae, which is measured from its post-YaRN base. Right, change in MATH500 when the Tulu3 SFT checkpoint of five bases receives 30K samples from one of four reasoning corpora, with 95% paired bootstrap intervals over the same 500 problems. Protocol B, single seed.

210 **Which subset carries the signal.** A mixture-composition ablation at a 100B-token budget isolates
 211 the two largest subsets of the permissive mixture (Tables 12 and 15). Removing the instruction-and-
 212 reasoning subset collapses the post-trained model to a suite average of 6 to 9 across contexts and
 213 stages, well below the full mixture, and Tulu3 does not repair it, which is the 100B counterpart of the
 214 front-loading effect in Section 3. Removing the curated web subset instead leaves both axes close
 215 to the full mixture and on several benchmarks above it, and costs about a point of mean accuracy
 216 on language understanding. At this budget the reasoning and instruction core carries most of the
 217 post-training signal and the web portion contributes comparatively little, though we state this only for
 218 the scale and budget tested.

219 5 Receptivity to reasoning post-training

220 Everything in this section is measured under Protocol B at a single seed, so every number below is a
 221 change from a starting checkpoint re-evaluated under that protocol rather than a score comparable
 222 with Sections 3 and 4.

223 **A small reasoning budget helps one base and harms the rest.** Under an identical 30K-sample
 224 OpenThoughts3 update, the front-loaded permissive base is the only model in our study whose
 225 suite average rises, by +3.6 points from its post-YaRN base (Figure 4, Table 10). Every other base
 226 declines from its own starting checkpoint, by −1.3 for Qwen2.5, −2.4 for FineWeb-Edu, −2.8
 227 for Nemotron-CC-HQ, −4.3 for SmolLM2, −5.0 for Comma0.1, and −19.3 for Qwen3. We read
 228 Qwen3’s decline as reduced susceptibility at a budget this small, since a recipe or budget mismatch
 229 would produce the same signature. The same update also recovers the long-context mathematics
 230 performance that Phase 1 erodes in Section 4, and does so from the base checkpoint without any
 231 Tulu3 stage, which supports reading that erosion as short-context interference.

232 **Removing the reasoning traces from pre-training removes the receptivity.** Table 3 isolates this
 233 at the full 300B budget. Two models differ only in whether the OpenThoughts3 subset is present
 234 in pre-training, and both receive identical reasoning post-training. With it the suite average reaches
 235 33.6, and without it the same post-training reaches 23.9, below the level the intact base already
 236 had before any reasoning update. The gap is 31.8 points on MATH500, 30.0 on AMC23, and 19.5
 237 on HumanEval, and AIME24 and AIME25 fall from 11.7 and 10.0 to exactly zero. Post-training
 238 amplified a capability that pre-training had laid down, and where it had not there was nothing to
 239 amplify.

240 **The effect is not an artifact of training twice on the same corpus.** Because OpenThoughts3
 241 traces are present in the permissive pre-training mixture, its gain under OpenThoughts3 post-training
 242 could reflect re-exposure rather than receptivity. Table 4 holds the intervention fixed and changes

Table 3: **Removing OpenThoughts3 from pre-training removes the receptivity to reasoning post-training.** Two 1.7B models pre-trained on 300B tokens that differ only in whether the OpenThoughts3 subset is present, then given the identical OT3-30K reasoning post-training. Architecture, schedule, token budget, long-context extension, post-training recipe, and evaluation are held fixed. The intact base is shown as a reference level, and the two post-trained endpoints are the comparison. Bold marks the condition of interest, the model whose pre-training retained OpenThoughts3. Protocol B, the reasoning post-training protocol (Appendix Table 27). Single seed, so no error bars are available.

300B model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Base with OT3, before reasoning post-training	3.3	8.3	25.6	34.6	54.1	42.1	8.4	29.2	27.3	10.2	45.2	26.2
OT3 in pre-training, → OT30K	11.7	10.0	47.5	69.8	56.4	57.9	12.7	26.6	28.3	20.6	27.6	33.6
OT3 removed from pre-training, → OT30K	0.0	0.0	17.5	38.0	53.3	38.4	6.2	33.8	28.8	11.0	36.4	23.9

Table 4: **The reasoning gain is not tied to OpenThoughts3.** Change in MATH500 accuracy when each base’s Tulu3 SFT checkpoint receives 30K samples from one of four reasoning corpora, three of which are absent from every pre-training mixture considered here. Brackets are 95% paired bootstrap intervals over the same 500 problems. **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction. Protocol B, the reasoning post-training protocol (Appendix Table 27). Single seed, so no error bars are available.

Base	OpenThoughts3	AceReason	Nemotron-Cascade	Superior-Reasoning
MixtureVitae	+35.4 [+30.4, +40.4]	+34.2 [+29.4, +38.8]	+31.6 [+27.0, +36.4]	+16.4 [+11.2, +21.4]
Comma0.1	−0.4 [−2.2, +1.4]	+0.4 [−1.4, +2.2]	+0.8 [−1.2, +2.8]	+0.0 [−1.6, +1.6]
FineWeb-Edu	−0.2 [−1.6, +1.2]	+1.0 [−0.4, +2.6]	+0.6 [−1.0, +2.2]	−0.8 [−2.0, +0.4]
Nemotron-CC-HQ	−2.2 [−3.8, −0.8]	+0.0 [−2.0, +2.0]	−0.8 [−2.6, +1.0]	−1.0 [−2.8, +0.8]
SmolLM2	−0.6 [−3.4, +2.2]	+8.4 [+5.0, +11.8]	+6.6 [+3.2, +10.2]	+5.6 [+2.2, +9.0]

only the identity of the reasoning corpus. From the Tulu3 SFT checkpoint the permissive base gains +35.4 points of MATH500 under OpenThoughts3 and +34.2, +31.6, and +16.4 under AceReason, Nemotron-Cascade, and Superior-Reasoning, three corpora that are not named sources in any pre-training mixture here, with every interval excluding zero. No matched-compute reference responds positively to any of the four, and SmolLM2, which is neither compute matched nor trained under our recipe, responds to three of the four roughly three to five times more weakly, and not at all to OpenThoughts3. Receptivity is therefore a property of the base rather than of the post-training corpus, although the corpus shapes the profile of the gain, with Superior-Reasoning improving mathematics while regressing code and instruction following enough to lower the average (Table 11).

What reasoning post-training costs. Instruction following is a control here rather than a target, and it declines under every reasoning update, since the post-training data contains none. Running DPO before the reasoning stage largely preserves it, at 43.0 against 45.0 before. General language understanding does not degrade, reading 56.0 after the reasoning update from either starting point against 55.0 through Phase 1 (Tables 16 and 19). More broadly the average over the non-mathematics benchmarks falls in most of these runs, so the gain from a small reasoning budget is real but concentrated, and is bought at a measurable cost in instruction following rather than in general language.

6 Related work

Permissive pre-training corpora. CommonCorpus [Langlais et al., 2026] assembles roughly 2T tokens of uncopyrighted and permissively licensed text, and CommonPile [Kandpal et al., 2025] builds pre-training corpora exclusively from permissively licensed sources. MixtureVitae [Nguyen et al., 2025] showed that much of the gap to mixed-license corpora closes at the base-model stage when a large instruction and reasoning fraction is included, and SYNTH [PleIAs, 2025] pursues a related synthetic route. None of these varies licensing and composition independently, which is what this paper adds.

Quality-filtered web corpora. FineWeb-Edu [Penedo et al., 2024] filters for educational value, and Nemotron-CC [Su et al., 2025] combines classifier ensembles with synthetic rephrasing. The

open-sci-ref protocol [Nezhurina et al., 2026] established the reference recipe we use and identified Nemotron-CC-HQ [Blakeman et al., 2025] as the strongest such corpus, ahead of DCLM [Li et al., 2024a] and FineWeb-Edu.

Instruction and reasoning data in pre-training. Phi-1 [Gunasekar et al., 2023] showed that textbook-quality and synthetic exercise data can carry a small model to strong code performance. Front-Loaded Reasoning [Akter et al., 2025] makes the pre-training-side case that reasoning data introduced early establishes capabilities that later supervised fine-tuning cannot fully replicate, and PRISM [Runwal et al., 2026] and Zhang et al. [2026] reach compatible conclusions from mid-training. Our design differs in three ways that matter for attribution. Pre-training here is single stage, so no stage boundary, schedule change, or curriculum shift co-varies with the mixture. Pre-training and all shared post-training data are decontaminated against every reported benchmark, which matters because front-loading mathematics and code is the regime where leakage is most likely. And licensing is varied rather than held implicitly fixed at non-permissive.

Post-training recipes. OpenThoughts [Guha et al., 2025] provides the reasoning traces and the recipe we use, and Tulu3 [Lambert et al., 2025] the instruction alignment recipe including DPO [Rafailov et al., 2023]. LIMA [Zhou et al., 2023a] and LIMO [Ye et al., 2025] argue that small post-training sets can elicit instruction following and mathematical reasoning from a well-pre-trained base, an amplification account they motivate but do not test by varying pre-training. A separate line induces reasoning through reinforcement learning with verifiable rewards or test-time scaling [Guo et al., 2025, Yu et al., 2025, Muennighoff et al., 2025]. Those papers vary the post-training algorithm or data. We hold post-training fixed and vary the pre-training mixture.

7 Conclusion and limitations

Asked what a base model must contain for post-training to succeed, this design answers with the pre-training mixture rather than the pre-training licence. Front-loading is worth 10 to 19 points of reasoning suite average in every cell of a 300B-token factorial, works equally well on permissive and non-permissive substrates, and helps broadly across mathematics, code, and instruction following. Under a permissive-data constraint the mechanism stays available essentially in full on the capabilities post-training targets and incomplete on general language coverage. For a community that wants to share data, recipes, and checkpoints without legal friction, that constraint is affordable, and composition rather than licensing is where the capability is decided.

Scale and substrate. All factorial results are at 1.7B parameters and 300B tokens, with a 0.4B cross-check in Appendix B.4, and the licensing factor has a single permissive level, so “permissive” here means MixtureVitae specifically. The same factorial at 7B and above, with CommonPile and CommonCorpus added as further levels, would turn a two-level contrast into a properly sampled factor and test whether the near-zero licensing penalty survives scale. This is the most valuable follow-up and the one we can least afford.

Post-training family. We test Tulu3 SFT and DPO and a 30K-sample reasoning update, without Tulu3’s reinforcement learning stage. Concluding that a weak mixture is genuinely unrecoverable would need exactly the regimes we omit, namely reinforcement learning with verifiable rewards and larger reasoning budgets, so we claim only that the deficit is not recovered under the recipes tested.

Measurement and coverage. Our intervals describe evaluation variability conditional on fixed checkpoints, so separating mixture effects from initialization variance needs replicate pre-training runs. The suite omits multilinguality, long-context retrieval, and tool use.

The language gap is open. Front-loading does not close the substrate gap on general language understanding, and whether targeted composition on a permissive substrate can is the most direct follow-up to the one negative result in this paper.

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Appendix

Broader impact

Our work provides evidence that permissively licensed pre-training data does not preclude strong post-training, which lowers the legal and practical barriers to sharing pre-training corpora, post-training recipes, and checkpoints. This favors transparency and reproducibility in a part of the field where both are scarce. The same finding makes capable models easier to build and to release, and the usual dual-use considerations for open model releases apply unchanged.

A Statistical methodology and full contrasts

What the intervals describe. Every Protocol A result is evaluated three times, with decoding seeds 42, 1234, and 2025, and reported as a mean with a sample standard deviation over those three runs. Under greedy decoding the residual run-to-run variation comes from batching and kernel nondeterminism in the inference engine rather than from sampling, which is why the standard deviations are small but nonzero, and occasionally large on the 30-problem contest-mathematics sets where a single problem is worth 3.3 points. These seeds do not estimate variability across independent pre-training runs. Doing so would require repeating the 300B-token pre-training runs, which was out of reach.

Paired item-level bootstrap. Contrasts on the suite average carry 95% percentile intervals from a paired item-level bootstrap with 2,000 replicates. Scores are averaged over the three seeds first, giving one score per item per checkpoint. Each replicate resamples items with replacement *within* each benchmark, which preserves the composition of the eleven-benchmark macro average in every replicate, and the contrast is recomputed on the resampled items. Because the two checkpoints of a contrast are evaluated on the same items, the resampling is paired and the interval reflects the uncertainty of the difference rather than of the two levels separately. JEEBench awards partial credit, so item scores lie in $[0, 1]$ and are treated as such rather than as Bernoulli outcomes. Two-sided p -values are taken from the bootstrap distribution and corrected within each contrast family with the Benjamini-Hochberg procedure. The three families are the front-loading effects, the substrate contrasts at matched front-loading, and the interaction, and Table 5 reports all of them.

Robustness to how the suite is weighted. The eleven-task macro average contains five mathematics and three code benchmarks, so it implicitly weights those domains at 45% and 27%. Table 6 recomputes the two headline quantities under an equal-domain average, which gives each of the four capability domains equal weight, and with each domain removed in turn. Neither the front-loading effect nor the substrate contrast depends on the weighting.

The contest-mathematics benchmarks. AIME24 and AIME25 contain 30 problems each and AMC23 contains 40. At those sizes a paired binomial power analysis puts the effect needed for 80% power at roughly 28 points for the AIME sets and 30 points for AMC23, so their percentage-point estimates are intrinsically imprecise and we do not build effect-size claims on them. Where we quote them, as in the pre-training ablation of Section 5 where AIME24 and AIME25 fall to exactly zero, the claim is that a model does or does not clear the benchmark floor, which is a qualitatively different and far more robust statement than an estimate of how far it clears it by.

B Additional results

B.1 The 300B factorial in full

Table 8 gives the per-benchmark values behind Section 3 for all thirty-six evaluated checkpoints, and Table 9 gives the language-understanding suite for the twelve distinct pre-training runs. Two patterns in the language table are worth naming. Without front-loading, FineWeb-Edu and the permissive substrate score at chance on both 5-shot MMLU and 10-shot CommonsenseQA, and front-loading moves them well above it, so part of what instruction and reasoning data supplies is the ability to answer a benchmark in the format it asks for. Nemotron-CC-HQ is the exception, retaining

Table 5: All Protocol A contrasts on the eleven-benchmark suite average, with 95% paired item-level bootstrap intervals (2,000 replicates, items resampled within benchmark, decoding seeds averaged first) and Benjamini–Hochberg adjusted q -values computed within each family. Positive values favour front-loading in the first block and the permissive substrate in the second. Estimates are computed on unrounded item-level scores, so they can differ in the final digit from the difference of the rounded values printed in Table 1.

Contrast	Stage	Estimate	95% CI	q
<i>Front-loading effect, within web substrate, 4K context</i>				
FineWeb-Edu	Base	+19.12	[+17.25, +20.88]	< 0.001
FineWeb-Edu	SFT	+15.57	[+14.31, +17.06]	< 0.001
FineWeb-Edu	DPO	+15.87	[+14.42, +17.36]	< 0.001
MixtureVitae	Base	+10.27	[+8.90, +11.67]	< 0.001
MixtureVitae	SFT	+11.82	[+10.27, +13.41]	< 0.001
MixtureVitae	DPO	+12.44	[+10.73, +14.20]	< 0.001
Nemotron-CC-HQ	Base	+19.17	[+17.38, +21.00]	< 0.001
Nemotron-CC-HQ	SFT	+14.45	[+13.05, +15.89]	< 0.001
Nemotron-CC-HQ	DPO	+15.23	[+13.81, +16.63]	< 0.001
<i>Front-loading effect, within web substrate, 16K context</i>				
FineWeb-Edu	Base	+19.06	[+17.43, +20.77]	< 0.001
FineWeb-Edu	SFT	+14.70	[+13.36, +16.08]	< 0.001
FineWeb-Edu	DPO	+13.63	[+12.23, +15.05]	< 0.001
MixtureVitae	Base	+19.09	[+17.34, +20.82]	< 0.001
MixtureVitae	SFT	+15.55	[+13.71, +17.42]	< 0.001
MixtureVitae	DPO	+15.76	[+13.94, +17.62]	< 0.001
Nemotron-CC-HQ	Base	+19.11	[+17.49, +20.83]	< 0.001
Nemotron-CC-HQ	SFT	+13.95	[+12.68, +15.29]	< 0.001
Nemotron-CC-HQ	DPO	+14.07	[+12.72, +15.48]	< 0.001
<i>Permissive minus non-permissive substrate, both front-loaded, 4K context</i>				
MV - FineWeb-Edu	Base	-7.49	[-9.30, -5.59]	< 0.001
MV - FineWeb-Edu	SFT	+0.34	[-1.28, +1.91]	0.656
MV - FineWeb-Edu	DPO	+0.94	[-0.90, +2.80]	0.429
MV - Nemotron	Base	-5.39	[-7.36, -3.41]	< 0.001
MV - Nemotron	SFT	+0.50	[-1.18, +2.26]	0.645
MV - Nemotron	DPO	+0.40	[-1.20, +2.01]	0.656
<i>Permissive minus non-permissive substrate, both front-loaded, 16K context</i>				
MV - FineWeb-Edu	Base	+3.54	[+1.75, +5.39]	0.002
MV - FineWeb-Edu	SFT	+4.77	[+3.04, +6.60]	< 0.001
MV - FineWeb-Edu	DPO	+5.66	[+3.94, +7.47]	< 0.001
MV - Nemotron	Base	+3.41	[+1.63, +5.10]	< 0.001
MV - Nemotron	SFT	+5.05	[+3.21, +6.96]	< 0.001
MV - Nemotron	DPO	+5.08	[+3.12, +7.01]	< 0.001
<i>Interaction: front-loading effect, permissive minus non-permissive, 4K context</i>				
MV - non-permissive	Base	-8.87	[-10.67, -7.01]	0.002
MV - non-permissive	SFT	-3.19	[-4.97, -1.46]	0.002
MV - non-permissive	DPO	-3.11	[-4.94, -1.20]	0.003
<i>Interaction: front-loading effect, permissive minus non-permissive, 16K context</i>				
MV - non-permissive	Base	+0.00	[-1.99, +1.95]	0.974
MV - non-permissive	SFT	+1.22	[-0.81, +3.23]	0.275
MV - non-permissive	DPO	+1.91	[-0.11, +3.98]	0.103

905 46.4 on MMLU and 48.6 on CommonsenseQA without front-loading, which is consistent with its
906 construction from classifier-filtered and synthetically rephrased web text and explains why it gains
907 least from the subset.

908 Figure 5 shows how the front-loading effect evolves across post-training stages. It is largest at the
909 base stage and shrinks under Phase 1 without disappearing, which is the same statement as the
910 observation in Section 3 that Tulu3 recovers only part of the deficit created by omitting the subset.
911 Figure 6 decomposes the effect by capability domain. Table 7 gives both context variants for every
912 model in the zoo, of which Table 2 in the main text shows one per family.

Table 6: Robustness of the two headline quantities to how the suite average is weighted, at 16K context after Tulu3 SFT. The eleven-task macro average implicitly weights math at 45% and code at 27%, and neither the front-loading effect nor the substrate contrast depends on that weighting. Protocol A (Appendix Table 26). Values are mean $\pm$ SD over decoding seeds 42, 1234, and 2025.

Averaging view	Front-loading effect			Permissive – non-permissive	
	MV	Nemotron	FineWeb	vs. Nemotron	vs. FineWeb
Suite average (11 tasks)	+15.5	+13.9	+14.7	+5.0	+4.8
Equal-domain average	+12.2	+12.8	+13.6	+3.2	+3.4
Drop math domain	+11.7	+15.0	+15.4	+3.7	+4.2
Drop code domain	+13.6	+10.3	+11.1	+3.6	+3.0
Drop science domain	+19.0	+16.1	+17.3	+6.5	+6.2
Drop instruction domain	+16.3	+14.3	+14.9	+5.7	+5.2

Table 7: Phase 1 (Tulu3 SFT and DPO) across the model zoo at 1.7B scale, both context variants. This is the complete version of Table 2. Protocol A (Appendix Table 26). Values are mean $\pm$ SD over decoding seeds 42, 1234, and 2025.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
<i>Matched compute, 300B tokens, controlled comparison</i>												
MixtureVitae	0.0 $\pm$ 0.00	7.8 $\pm$ 5.09	5.0 $\pm$ 0.00	17.8 $\pm$ 0.72	54.1 $\pm$ 0.00	13.6 $\pm$ 0.93	13.1 $\pm$ 1.36	19.5 $\pm$ 0.12	24.4 $\pm$ 0.77	2.9 $\pm$ 0.51	36.2 $\pm$ 0.13	17.7 $\pm$ 0.29
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	12.5 $\pm$ 0.00	29.5 $\pm$ 0.23	49.9 $\pm$ 0.00	44.3 $\pm$ 0.70	2.4 $\pm$ 0.11	39.9 $\pm$ 0.23	20.7 $\pm$ 0.87	3.7 $\pm$ 0.55	49.8 $\pm$ 0.13	23.0 $\pm$ 0.16
→ SFT → DPO	0.0 $\pm$ 0.00	3.3 $\pm$ 0.00	10.8 $\pm$ 1.44	30.5 $\pm$ 0.12	50.2 $\pm$ 0.00	47.4 $\pm$ 0.35	2.0 $\pm$ 0.20	41.2 $\pm$ 0.20	21.2 $\pm$ 0.00	4.5 $\pm$ 0.34	52.2 $\pm$ 0.20	23.9 $\pm$ 0.17
MixtureVitae-16K	1.1 $\pm$ 1.92	3.3 $\pm$ 0.00	23.3 $\pm$ 8.04	47.0 $\pm$ 0.53	60.0 $\pm$ 0.00	54.1 $\pm$ 0.35	22.3 $\pm$ 0.34	40.1 $\pm$ 0.23	23.4 $\pm$ 0.29	8.0 $\pm$ 0.24	54.6 $\pm$ 0.23	30.7 $\pm$ 0.72
→ SFT	3.3 $\pm$ 0.00	3.3 $\pm$ 0.00	15.0 $\pm$ 0.00	35.0 $\pm$ 0.53	51.6 $\pm$ 0.00	54.1 $\pm$ 0.35	14.4 $\pm$ 0.11	40.4 $\pm$ 0.35	23.4 $\pm$ 0.29	5.1 $\pm$ 0.27	55.1 $\pm$ 0.26	27.3 $\pm$ 0.11
→ SFT → DPO	3.3 $\pm$ 0.00	1.1 $\pm$ 1.92	17.5 $\pm$ 0.00	37.0 $\pm$ 0.80	52.6 $\pm$ 0.00	52.6 $\pm$ 0.70	14.6 $\pm$ 0.30	40.3 $\pm$ 0.12	23.4 $\pm$ 0.29	6.2 $\pm$ 0.20	56.5 $\pm$ 0.74	27.8 $\pm$ 0.04
Nemotron-CC-HQ	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.00	4.1 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.4 $\pm$ 0.00	5.7 $\pm$ 0.29	0.0 $\pm$ 0.00	30.3 $\pm$ 0.20	3.9 $\pm$ 0.01
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	5.0 $\pm$ 0.00	3.1 $\pm$ 0.23	6.8 $\pm$ 0.00	4.9 $\pm$ 0.00	0.2 $\pm$ 0.00	5.5 $\pm$ 0.23	20.2 $\pm$ 0.00	2.0 $\pm$ 0.17	40.5 $\pm$ 0.27	8.0 $\pm$ 0.04
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	3.5 $\pm$ 0.42	8.0 $\pm$ 0.00	6.1 $\pm$ 0.00	0.2 $\pm$ 0.00	5.1 $\pm$ 0.12	21.2 $\pm$ 0.00	2.7 $\pm$ 0.66	42.1 $\pm$ 0.39	8.3 $\pm$ 0.07
Nemotron-CC-HQ-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.8 $\pm$ 0.35	8.2 $\pm$ 0.00	1.8 $\pm$ 0.00	0.1 $\pm$ 0.11	5.7 $\pm$ 0.31	22.7 $\pm$ 0.51	5.6 $\pm$ 1.04	42.7 $\pm$ 0.38	8.1 $\pm$ 0.08
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.9 $\pm$ 0.31	8.6 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.9 $\pm$ 0.12	22.2 $\pm$ 0.87	1.2 $\pm$ 0.08	45.8 $\pm$ 0.00	8.4 $\pm$ 0.12
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.5 $\pm$ 0.12	7.7 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.3 $\pm$ 0.12	23.4 $\pm$ 1.17	1.7 $\pm$ 0.37	46.3 $\pm$ 0.61	8.6 $\pm$ 0.06
FineWeb-Edu	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.2 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	25.1 $\pm$ 0.58	0.0 $\pm$ 0.00	38.1 $\pm$ 0.08	6.0 $\pm$ 0.06
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.1 $\pm$ 0.12	7.1 $\pm$ 0.00	2.4 $\pm$ 0.00	0.0 $\pm$ 0.00	2.3 $\pm$ 0.23	26.9 $\pm$ 0.58	1.6 $\pm$ 0.46	35.3 $\pm$ 0.15	7.1 $\pm$ 0.07
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	1.7 $\pm$ 0.31	6.7 $\pm$ 0.00	2.4 $\pm$ 0.00	0.0 $\pm$ 0.00	1.9 $\pm$ 0.12	25.9 $\pm$ 0.29	1.6 $\pm$ 0.17	35.5 $\pm$ 0.13	7.1 $\pm$ 0.07
FineWeb-Edu-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.9 $\pm$ 0.12	8.6 $\pm$ 0.00	2.4 $\pm$ 0.00	0.1 $\pm$ 0.11	2.0 $\pm$ 0.00	24.9 $\pm$ 0.29	6.5 $\pm$ 0.58	38.8 $\pm$ 0.53	8.1 $\pm$ 0.09
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.1 $\pm$ 0.12	8.7 $\pm$ 0.00	4.9 $\pm$ 0.00	0.2 $\pm$ 0.00	2.5 $\pm$ 0.23	23.6 $\pm$ 0.58	2.4 $\pm$ 0.40	41.3 $\pm$ 0.46	7.9 $\pm$ 0.07
→ SFT → DPO	0.0 $\pm$ 0.00	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	3.3 $\pm$ 0.12	9.2 $\pm$ 0.00	4.9 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.12	23.9 $\pm$ 0.29	2.7 $\pm$ 0.54	43.3 $\pm$ 0.57	8.5 $\pm$ 0.09
DCLM	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.00	3.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	19.5 $\pm$ 0.77	0.0 $\pm$ 0.00	39.2 $\pm$ 0.08	5.6 $\pm$ 0.08
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.8 $\pm$ 0.00	5.2 $\pm$ 0.00	6.7 $\pm$ 0.00	0.2 $\pm$ 0.00	5.4 $\pm$ 0.35	21.7 $\pm$ 0.00	1.1 $\pm$ 0.49	40.6 $\pm$ 0.13	7.4 $\pm$ 0.02
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	1.1 $\pm$ 0.23	4.6 $\pm$ 0.00	6.7 $\pm$ 0.00	0.3 $\pm$ 0.11	5.6 $\pm$ 0.00	20.7 $\pm$ 0.00	0.5 $\pm$ 0.03	43.0 $\pm$ 0.08	7.7 $\pm$ 0.15
Comma0.1	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	5.5 $\pm$ 0.00	7.9 $\pm$ 0.00	0.0 $\pm$ 0.00	8.8 $\pm$ 0.20	20.2 $\pm$ 1.34	0.0 $\pm$ 0.00	39.0 $\pm$ 0.00	7.4 $\pm$ 0.12
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.23	9.3 $\pm$ 0.00	15.2 $\pm$ 0.00	1.8 $\pm$ 0.00	22.5 $\pm$ 0.46	25.6 $\pm$ 0.29	3.9 $\pm$ 0.25	44.6 $\pm$ 0.13	11.4 $\pm$ 0.05
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.31	9.7 $\pm$ 0.00	15.2 $\pm$ 0.00	1.8 $\pm$ 0.00	22.7 $\pm$ 0.12	26.8 $\pm$ 1.34	3.9 $\pm$ 0.49	46.9 $\pm$ 0.20	11.8 $\pm$ 0.09
Comma0.1-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	2.8 $\pm$ 0.35	12.2 $\pm$ 0.00	13.4 $\pm$ 0.00	0.1 $\pm$ 0.11	16.0 $\pm$ 0.20	25.8 $\pm$ 1.01	7.8 $\pm$ 0.19	43.8 $\pm$ 0.81	11.2 $\pm$ 0.22
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.5 $\pm$ 0.12	6.9 $\pm$ 0.00	16.5 $\pm$ 0.00	1.4 $\pm$ 0.20	25.2 $\pm$ 0.35	23.2 $\pm$ 0.00	4.8 $\pm$ 0.66	49.0 $\pm$ 0.27	12.0 $\pm$ 0.01
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.1 $\pm$ 0.12	6.7 $\pm$ 0.00	17.1 $\pm$ 0.00	1.8 $\pm$ 0.23	25.9 $\pm$ 0.31	23.4 $\pm$ 0.29	5.1 $\pm$ 0.12	50.6 $\pm$ 0.23	12.3 $\pm$ 0.03
<i>External open-weights references, 11–36T tokens, uncontrolled</i>												
SmolLM2	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	29.3 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.9 $\pm$ 0.12	15.7 $\pm$ 1.01	0.0 $\pm$ 0.00	32.4 $\pm$ 0.08	7.2 $\pm$ 0.09
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 2.50	6.3 $\pm$ 0.12	33.3 $\pm$ 0.00	30.9 $\pm$ 0.70	5.3 $\pm$ 0.20	37.2 $\pm$ 0.20	24.7 $\pm$ 1.01	3.7 $\pm$ 0.37	52.3 $\pm$ 0.40	17.8 $\pm$ 0.33
→ SFT → DPO	1.1 $\pm$ 1.92	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	8.3 $\pm$ 0.46	33.0 $\pm$ 0.00	30.9 $\pm$ 1.76	5.0 $\pm$ 0.11	37.7 $\pm$ 0.42	25.8 $\pm$ 0.87	4.4 $\pm$ 0.30	53.4 $\pm$ 0.34	18.3 $\pm$ 0.18
SmolLM2-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.8 $\pm$ 1.44	11.8 $\pm$ 0.92	31.9 $\pm$ 0.00	24.0 $\pm$ 0.35	1.2 $\pm$ 0.23	36.5 $\pm$ 0.12	24.6 $\pm$ 0.29	7.6 $\pm$ 0.39	49.9 $\pm$ 0.27	17.1 $\pm$ 0.07
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	1.7 $\pm$ 1.44	9.6 $\pm$ 0.20	32.8 $\pm$ 0.00	32.9 $\pm$ 0.61	4.4 $\pm$ 0.11	37.5 $\pm$ 0.23	24.9 $\pm$ 1.91	4.7 $\pm$ 0.72	54.4 $\pm$ 0.40	18.4 $\pm$ 0.05
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.3 $\pm$ 1.44	9.6 $\pm$ 0.53	33.8 $\pm$ 0.00	32.3 $\pm$ 1.06	5.3 $\pm$ 0.20	36.9 $\pm$ 0.50	26.3 $\pm$ 0.00	5.4 $\pm$ 0.69	57.4 $\pm$ 0.20	19.1 $\pm$ 0.17
Qwen2.5	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.35	61.9 $\pm$ 0.00	5.7 $\pm$ 0.70	0.2 $\pm$ 0.00	0.1 $\pm$ 0.12	11.1 $\pm$ 1.82	0.3 $\pm$ 0.10	31.3 $\pm$ 0.34	10.1 $\pm$ 0.15
→ SFT	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	8.3 $\pm$ 1.44	18.8 $\pm$ 0.69	39.0 $\pm$ 0.00	51.0 $\pm$ 0.35	8.0 $\pm$ 0.23	44.3 $\pm$ 0.23	29.0 $\pm$ 0.77	6.1 $\pm$ 0.54	56.0 $\pm$ 0.27	24.0 $\pm$ 0.07
→ SFT → DPO	2.2 $\pm$ 1.92	0.0 $\pm$ 0.00	10.0 $\pm$ 2.50	21.2 $\pm$ 1.00	40.4 $\pm$ 0.00	55.7 $\pm$ 0.70	8.5 $\pm$ 0.23	44.5 $\pm$ 0.23	27.9 $\pm$ 0.58	8.0 $\pm$ 0.64	61.2 $\pm$ 0.30	25.4 $\pm$ 0.40
Qwen3	3.3 $\pm$ 0.00	2.2 $\pm$ 1.92	20.8 $\pm$ 3.82	48.0 $\pm$ 0.35	71.6 $\pm$ 0.00	64.2 $\pm$ 0.93	1.6 $\pm$ 0.11	46.5 $\pm$ 0.81	25.1 $\pm$ 0.77	18.0 $\pm$ 0.76	38.8 $\pm$ 0.34	30.9 $\pm$ 0.31
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	16.7 $\pm$ 2.89	28.2 $\pm$ 0.35	69.4 $\pm$ 0.00	67.1 $\pm$ 0.61	3.5 $\pm$ 0.85	54.3 $\pm$ 0.58	33.0 $\pm$ 1.05	11.9 $\pm$ 0.54	63.7 $\pm$ 0.13	31.6 $\pm$ 0.30
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	8.3 $\pm$ 1.44	31.2 $\pm$ 1.06	69.8 $\pm$ 0.00	67.7 $\pm$ 0.61	4.9 $\pm$ 0.00	54.7 $\pm$ 0.23	31.3 $\pm$ 0.87	13.3 $\pm$ 1.25	67.6 $\pm$ 1.33	31.7 $\pm$ 0.21

913 B.2 Reasoning post-training endpoints

914 Table 10 gives the per-benchmark endpoints behind Figure 4, with each block’s first row the starting
915 checkpoint that the OpenThoughts3-30K run was initialised from and re-evaluated under Protocol B,
916 so that every comparison inside the table is paired. Table 11 gives the full held-out reasoning grid,
917 four reasoning corpora applied from three starting checkpoints with recipe, sample budget, and epoch
918 count held fixed. Two observations that the compact main-text version omits. Superior-Reasoning
919 is a mixed control rather than a failure, improving mathematics substantially while regressing code
920 and instruction following enough to lower the suite average from every starting point. And the DPO
921 starting point is the one that preserves instruction following through reasoning post-training, ending
922 at 43.0 on IFEval against 45.0 before the update, where the SFT starting point falls to 25.6.

Table 8: Per-benchmark results for the 300B front-loading $\times$ web-substrate factorial. (FL) marks the models whose pre-training mixture includes the MixtureVitae instruction-and-reasoning subset. Protocol A (Appendix Table 26). Values are mean $\pm$ SD over decoding seeds 42, 1234, and 2025.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
MixtureVitae (FL)	0.0 $\pm$ 0.00	7.8 $\pm$ 5.09	5.0 $\pm$ 0.00	17.8 $\pm$ 0.72	54.1 $\pm$ 0.00	13.6 $\pm$ 0.93	13.1 $\pm$ 1.36	19.5 $\pm$ 0.12	24.4 $\pm$ 0.77	2.9 $\pm$ 0.51	36.2 $\pm$ 0.13	17.7 $\pm$ 0.29
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	12.5 $\pm$ 0.00	29.5 $\pm$ 0.23	49.9 $\pm$ 0.00	44.3 $\pm$ 0.70	2.4 $\pm$ 0.11	39.9 $\pm$ 0.23	20.7 $\pm$ 0.87	3.7 $\pm$ 0.55	49.8 $\pm$ 0.13	23.0 $\pm$ 0.16
→ SFT → DPO	0.0 $\pm$ 0.00	3.3 $\pm$ 0.00	10.8 $\pm$ 1.44	30.5 $\pm$ 0.12	50.2 $\pm$ 0.00	47.4 $\pm$ 0.35	2.0 $\pm$ 0.20	41.2 $\pm$ 0.20	21.2 $\pm$ 0.00	4.5 $\pm$ 0.34	52.2 $\pm$ 0.20	23.9 $\pm$ 0.17
MixtureVitae	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.1 $\pm$ 0.12	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	22.0 $\pm$ 0.20	23.6 $\pm$ 0.29	0.0 $\pm$ 0.00	32.3 $\pm$ 0.20	7.4 $\pm$ 0.02
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	1.9 $\pm$ 0.12	4.4 $\pm$ 0.00	19.1 $\pm$ 0.35	1.8 $\pm$ 0.00	25.1 $\pm$ 0.50	25.1 $\pm$ 0.29	2.3 $\pm$ 0.30	40.4 $\pm$ 0.13	11.1 $\pm$ 0.03
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	1.7 $\pm$ 0.23	4.0 $\pm$ 0.00	19.7 $\pm$ 0.35	1.8 $\pm$ 0.00	25.4 $\pm$ 0.00	25.1 $\pm$ 0.29	2.5 $\pm$ 0.53	43.7 $\pm$ 0.00	11.5 $\pm$ 0.09
Nemotron (FL)	1.1 $\pm$ 1.92	5.6 $\pm$ 1.92	25.0 $\pm$ 0.00	44.6 $\pm$ 0.69	52.7 $\pm$ 0.00	40.2 $\pm$ 0.61	16.4 $\pm$ 0.20	8.7 $\pm$ 0.12	20.4 $\pm$ 1.27	3.3 $\pm$ 0.42	35.6 $\pm$ 0.30	23.1 $\pm$ 0.33
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	7.5 $\pm$ 0.00	19.4 $\pm$ 0.40	50.9 $\pm$ 0.00	47.8 $\pm$ 0.70	6.9 $\pm$ 0.11	35.1 $\pm$ 0.12	24.2 $\pm$ 0.00	4.3 $\pm$ 0.15	51.0 $\pm$ 0.45	22.5 $\pm$ 0.10
→ SFT → DPO	1.1 $\pm$ 1.92	0.0 $\pm$ 0.00	7.5 $\pm$ 2.50	19.9 $\pm$ 0.99	51.5 $\pm$ 0.00	50.0 $\pm$ 0.00	8.5 $\pm$ 0.30	35.5 $\pm$ 0.23	26.1 $\pm$ 0.29	4.8 $\pm$ 0.78	54.1 $\pm$ 0.90	23.5 $\pm$ 0.21
Nemotron	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.2 $\pm$ 0.00	4.1 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.4 $\pm$ 0.00	5.7 $\pm$ 0.29	0.0 $\pm$ 0.00	30.3 $\pm$ 0.20	3.9 $\pm$ 0.01
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	5.0 $\pm$ 0.00	3.1 $\pm$ 0.23	6.8 $\pm$ 0.00	4.9 $\pm$ 0.00	0.2 $\pm$ 0.00	5.5 $\pm$ 0.23	20.2 $\pm$ 0.00	2.0 $\pm$ 0.17	40.5 $\pm$ 0.27	8.0 $\pm$ 0.04
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	3.5 $\pm$ 0.42	8.0 $\pm$ 0.00	6.1 $\pm$ 0.00	0.2 $\pm$ 0.00	5.1 $\pm$ 0.12	21.2 $\pm$ 0.00	2.7 $\pm$ 0.66	42.1 $\pm$ 0.39	8.3 $\pm$ 0.07
FineWeb-Edu (FL)	0.0 $\pm$ 0.00	6.7 $\pm$ 0.00	22.5 $\pm$ 0.00	47.0 $\pm$ 1.11	53.5 $\pm$ 0.00	38.4 $\pm$ 0.00	16.0 $\pm$ 0.30	28.5 $\pm$ 0.64	23.1 $\pm$ 0.29	5.0 $\pm$ 0.39	36.0 $\pm$ 0.13	25.2 $\pm$ 0.07
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	9.2 $\pm$ 1.44	17.9 $\pm$ 0.42	50.1 $\pm$ 0.00	51.2 $\pm$ 0.00	6.7 $\pm$ 0.23	34.4 $\pm$ 0.20	23.6 $\pm$ 0.29	4.4 $\pm$ 0.14	51.4 $\pm$ 0.60	22.6 $\pm$ 0.15
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	7.5 $\pm$ 0.00	18.4 $\pm$ 0.20	50.2 $\pm$ 0.00	53.0 $\pm$ 0.00	7.4 $\pm$ 0.20	35.2 $\pm$ 0.00	23.7 $\pm$ 0.00	4.4 $\pm$ 0.18	53.0 $\pm$ 0.49	23.0 $\pm$ 0.08
FineWeb-Edu	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.2 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	25.1 $\pm$ 0.58	0.0 $\pm$ 0.00	38.1 $\pm$ 0.08	6.0 $\pm$ 0.06
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.1 $\pm$ 0.12	7.1 $\pm$ 0.00	2.4 $\pm$ 0.00	0.0 $\pm$ 0.00	2.3 $\pm$ 0.23	26.9 $\pm$ 0.58	1.6 $\pm$ 0.46	35.3 $\pm$ 0.15	7.1 $\pm$ 0.07
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	1.7 $\pm$ 0.31	6.7 $\pm$ 0.00	2.4 $\pm$ 0.00	0.0 $\pm$ 0.00	1.9 $\pm$ 0.12	25.9 $\pm$ 0.29	1.6 $\pm$ 0.17	35.5 $\pm$ 0.13	7.1 $\pm$ 0.07
MixtureVitae-16K (FL)	1.1 $\pm$ 1.92	3.3 $\pm$ 0.00	23.3 $\pm$ 8.04	47.0 $\pm$ 0.53	60.0 $\pm$ 0.00	54.1 $\pm$ 0.35	22.3 $\pm$ 0.34	40.1 $\pm$ 0.23	23.4 $\pm$ 0.29	8.0 $\pm$ 0.24	54.6 $\pm$ 0.23	30.7 $\pm$ 0.72
→ SFT	3.3 $\pm$ 0.00	3.3 $\pm$ 0.00	15.0 $\pm$ 0.00	35.0 $\pm$ 0.53	51.6 $\pm$ 0.00	54.1 $\pm$ 0.35	14.4 $\pm$ 0.11	40.4 $\pm$ 0.35	23.4 $\pm$ 0.29	5.1 $\pm$ 0.27	55.1 $\pm$ 0.26	27.3 $\pm$ 0.11
→ SFT → DPO	3.3 $\pm$ 0.00	1.1 $\pm$ 1.92	17.5 $\pm$ 0.00	37.0 $\pm$ 0.80	52.6 $\pm$ 0.00	52.6 $\pm$ 0.70	14.6 $\pm$ 0.30	40.3 $\pm$ 0.12	23.4 $\pm$ 0.29	6.2 $\pm$ 0.20	56.5 $\pm$ 0.74	27.8 $\pm$ 0.04
MixtureVitae-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.6 $\pm$ 0.20	5.4 $\pm$ 0.00	19.3 $\pm$ 0.35	0.0 $\pm$ 0.00	22.3 $\pm$ 0.23	23.2 $\pm$ 0.00	6.5 $\pm$ 0.07	45.4 $\pm$ 0.20	11.6 $\pm$ 0.03
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.3 $\pm$ 0.12	5.0 $\pm$ 0.00	19.5 $\pm$ 0.00	1.3 $\pm$ 0.11	25.9 $\pm$ 0.42	25.1 $\pm$ 0.29	3.4 $\pm$ 0.18	47.2 $\pm$ 0.30	11.8 $\pm$ 0.02
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.8 $\pm$ 1.44	2.6 $\pm$ 0.20	4.6 $\pm$ 0.00	19.1 $\pm$ 0.35	1.3 $\pm$ 0.11	25.6 $\pm$ 0.00	25.1 $\pm$ 0.58	3.9 $\pm$ 0.53	48.8 $\pm$ 0.38	12.0 $\pm$ 0.15
Nemotron-16K (FL)	0.0 $\pm$ 0.00	1.1 $\pm$ 1.92	20.0 $\pm$ 2.50	39.9 $\pm$ 0.50	59.8 $\pm$ 0.00	46.5 $\pm$ 0.35	11.7 $\pm$ 0.11	34.7 $\pm$ 0.12	27.3 $\pm$ 0.00	8.3 $\pm$ 0.83	50.4 $\pm$ 0.40	27.3 $\pm$ 0.38
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	6.7 $\pm$ 1.44	18.5 $\pm$ 0.46	49.6 $\pm$ 0.00	49.8 $\pm$ 0.35	0.2 $\pm$ 0.00	32.5 $\pm$ 0.23	25.9 $\pm$ 0.29	6.0 $\pm$ 0.29	56.1 $\pm$ 0.30	22.3 $\pm$ 0.27
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	5.8 $\pm$ 1.44	20.3 $\pm$ 0.50	50.7 $\pm$ 0.00	49.4 $\pm$ 0.61	0.4 $\pm$ 0.00	32.9 $\pm$ 0.12	24.7 $\pm$ 0.00	6.8 $\pm$ 0.50	58.3 $\pm$ 0.72	22.7 $\pm$ 0.27
Nemotron-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.8 $\pm$ 0.35	8.2 $\pm$ 0.00	1.8 $\pm$ 0.00	0.1 $\pm$ 0.11	5.7 $\pm$ 0.31	22.7 $\pm$ 0.51	5.6 $\pm$ 1.04	42.7 $\pm$ 0.38	8.1 $\pm$ 0.08
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.9 $\pm$ 0.31	8.6 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.9 $\pm$ 0.12	22.2 $\pm$ 0.87	1.2 $\pm$ 0.08	45.8 $\pm$ 0.00	8.4 $\pm$ 0.12
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.5 $\pm$ 0.12	7.7 $\pm$ 0.00	5.3 $\pm$ 0.35	0.0 $\pm$ 0.00	5.3 $\pm$ 0.12	23.4 $\pm$ 1.17	1.7 $\pm$ 0.37	46.3 $\pm$ 0.61	8.6 $\pm$ 0.06
FineWeb-Edu-16K (FL)	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	13.3 $\pm$ 1.44	38.0 $\pm$ 0.35	57.2 $\pm$ 0.00	43.9 $\pm$ 0.00	18.1 $\pm$ 0.30	35.3 $\pm$ 0.12	29.0 $\pm$ 0.29	9.1 $\pm$ 0.97	54.5 $\pm$ 0.45	27.1 $\pm$ 0.06
→ SFT	0.0 $\pm$ 0.00	2.2 $\pm$ 1.92	5.8 $\pm$ 1.44	21.9 $\pm$ 0.64	51.1 $\pm$ 0.00	45.5 $\pm$ 0.35	3.5 $\pm$ 0.00	31.7 $\pm$ 0.23	26.8 $\pm$ 0.51	5.4 $\pm$ 0.30	54.4 $\pm$ 0.23	22.6 $\pm$ 0.24
→ SFT → DPO	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	23.1 $\pm$ 0.31	51.1 $\pm$ 0.00	45.1 $\pm$ 0.61	3.5 $\pm$ 0.00	31.4 $\pm$ 0.00	25.3 $\pm$ 0.00	5.1 $\pm$ 1.05	55.9 $\pm$ 0.00	22.1 $\pm$ 0.07
FineWeb-Edu-16K	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.00	2.9 $\pm$ 0.12	8.6 $\pm$ 0.00	2.4 $\pm$ 0.00	0.1 $\pm$ 0.11	2.0 $\pm$ 0.00	24.9 $\pm$ 0.29	6.5 $\pm$ 0.58	38.8 $\pm$ 0.53	8.1 $\pm$ 0.09
→ SFT	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	0.0 $\pm$ 0.00	3.1 $\pm$ 0.12	8.7 $\pm$ 0.00	4.9 $\pm$ 0.00	0.2 $\pm$ 0.00	2.5 $\pm$ 0.23	23.0 $\pm$ 0.58	2.4 $\pm$ 0.40	41.3 $\pm$ 0.46	7.9 $\pm$ 0.07
→ SFT → DPO	0.0 $\pm$ 0.00	3.3 $\pm$ 0.00	0.0 $\pm$ 0.00	3.3 $\pm$ 0.12	9.2 $\pm$ 0.00	4.9 $\pm$ 0.00	0.0 $\pm$ 0.00	2.5 $\pm$ 0.12	23.9 $\pm$ 0.29	2.7 $\pm$ 0.54	43.3 $\pm$ 0.57	8.5 $\pm$ 0.09

Table 9: Language understanding for the twelve factorial checkpoints, on the base and post-YaRN 16K checkpoints. This suite is scored by the LM Evaluation Harness under its own settings and is not on either reasoning protocol. Single seed. Accuracies in percent, acc_norm where the harness reports it, and shot counts are in Table 28. Without front-loading, FineWeb-Edu and the MixtureVitae substrate sit at chance on MMLU and CommonsenseQA.

Model	FL	MMLU	HellaS	CSQA	ARC-C	ARC-E	PIQA	BoolQ	WinoG	OBQA	COPA	LAMBADA	Avg
MixtureVitae	yes	38.7	54.6	48.7	41.0	72.4	70.7	73.3	59.1	36.2	72.0	49.2	56.0
MixtureVitae	no	25.8	52.2	22.4	38.2	68.8	70.5	64.8	58.2	36.4	73.0	51.2	51.0
Nemotron	yes	50.5	68.9	62.1	52.3	80.3	77.6	79.3	62.9	38.6	80.0	57.7	64.6
Nemotron	no	46.4	71.5	48.6	52.4	80.9	78.7	77.4	65.0	42.0	85.0	58.0	64.2
FineWeb-Edu	yes	47.5	65.3	55.7	49.5	79.9	76.9	77.3	62.6	43.0	79.0	54.6	62.8
FineWeb-Edu	no	26.1	67.1	21.1	48.5	76.9	77.1	64.9	62.4	42.4	82.0	55.7	56.7
MixtureVitae-16K	yes	38.1	52.9	47.7	41.1	73.0	69.6	75.1	58.6	36.8	71.0	50.4	55.8
MixtureVitae-16K	no	26.1	51.0	22.5	38.9	68.6	69.9	61.9	57.9	34.6	76.0	51.8	50.8
Nemotron-16K	yes	49.5	67.7	61.7	52.3	81.4	77.9	79.5	62.8	38.2	81.0	58.5	64.6
Nemotron-16K	no	46.5	70.0	50.6	52.0	80.7	78.1	77.1	64.6	41.8	83.0	60.1	64.1
FineWeb-Edu-16K	yes	46.4	62.5	56.4	50.2	79.6	76.7	77.6	62.1	42.0	79.0	56.1	62.6
FineWeb-Edu-16K	no	26.3	64.6	21.0	48.4	78.8	76.6	70.5	62.9	42.8	82.0	57.4	57.4

923 B.3 Mixture composition at a 100B budget

924 Tables 12 and 15 give the per-benchmark values behind the mixture-composition ablation of Sec-
 925 tion 4, in which two ablated variants of the permissive mixture are pre-trained at 100B tokens and
 926 passed through the same Phase 1 pipeline. Removing the instruction-and-reasoning subset collapses
 927 performance across mathematics, code, and reasoning at both context lengths, with AIME24 and
 928 AIME25 at zero, MATH500 below three, and suite averages of 6.1 to 8.5 at 4K and 6.7 to 7.5 at
 929 16K, and Tulu3 SFT does not repair it. Removing the curated web subset behaves very differently,
 930 reaching 18.6 at the 4K base and 16.1 to 17.5 after Phase 1, never below the full mixture and above
 931 it on MATH500, AMC23, HumanEval, and MBPP. On language understanding the web ablation
 932 costs about a point, 49.0 to 50.0 mean accuracy against 50.0 to 51.0 for the full mixture, while the

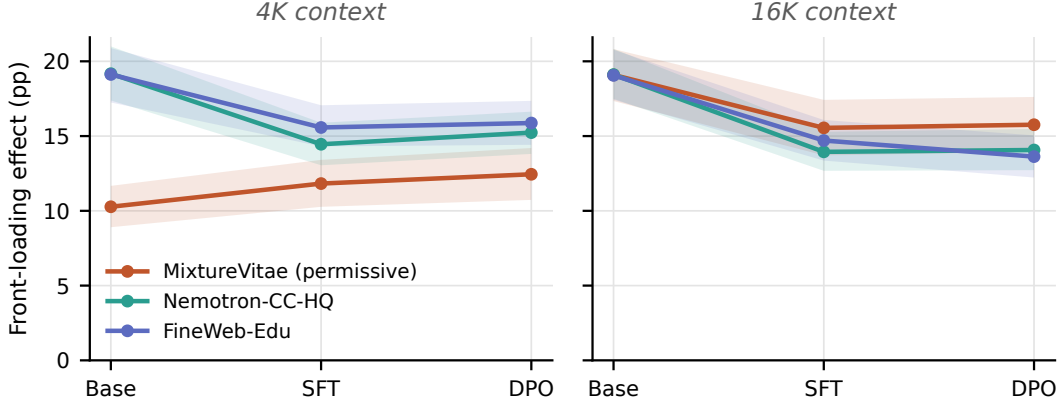


Figure 5: Front-loading effect on the reasoning suite average across post-training stages, with 95% paired item-level bootstrap intervals. Protocol A.

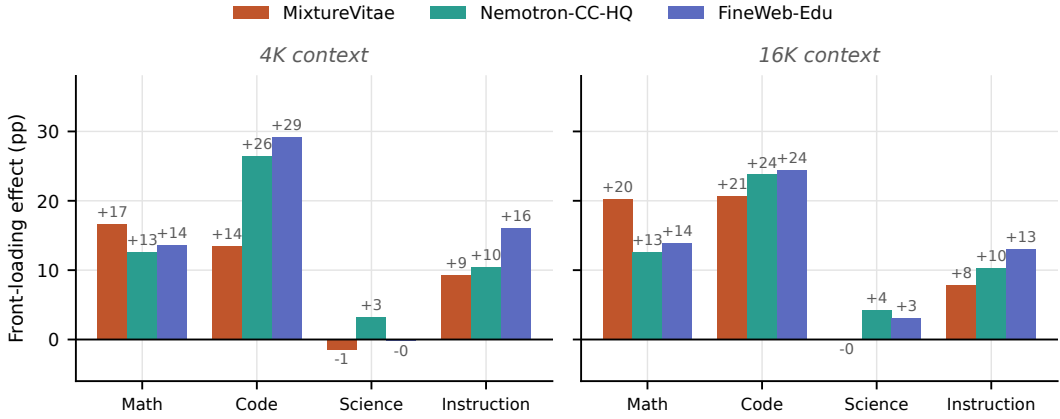


Figure 6: Front-loading effect by capability domain after Tulu3 SFT. The gain is broad rather than confined to mathematics, and instruction following improves for every substrate. Protocol A.

933 instruction ablation costs 4 to 5 points that Phase 1 does not close. We state this for the scale and
 934 budget tested and do not generalise it.

935 B.4 Smaller scale check at 0.4B

936 Tables 14 and 18 compare a 0.4B permissive model against Baguettotron, a 0.32B model trained on
 937 the PleIAs SYNTH mix [PleIAs, 2025], under the same Phase 1 pipeline. The permissive model
 938 reaches 12.1 reasoning suite average after DPO against 6.2 for Baguettotron, with HumanEval and
 939 MBPP pass@1 of 15.3 and 22.6 against near zero, and 45.0 against 42.0 to 43.0 mean accuracy on
 940 the language suite. The decontamination status of the SYNTH mix is undocumented while our model
 941 is decontaminated against every benchmark reported here, so some part of Baguettotron’s score may
 942 reflect train and test overlap that we cannot rule out from outside the release. This comparison is a
 943 sanity check on the direction of the effect at a second scale and not a scaling result.

944 C Experimental setup details

945 C.1 open-sci-ref setup and hyperparameters

946 This appendix details the exact model architectures and training hyperparameters used for all
 947 open-sci-ref-based pre-training.

Table 10: Reasoning post-training endpoints. Identical OpenThoughts3-30K post-training (30K samples, 5 epochs) is applied to every model, starting from the checkpoint named in each block’s first row. Starting checkpoints are re-evaluated here, so these values are paired within the table and must not be compared against the Phase 1 tables. Protocol B, the reasoning post-training protocol (Appendix Table 27). Single seed, so no error bars are available.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
MV-16k (base)	0.0	10.0	30.0	33.6	58.4	48.2	11.7	41.0	26.3	8.3	42.6	28.2
→ SFT → OT30K	20.0	16.7	29.8	72.8	52.5	58.5	10.0	29.2	36.4	4.6	25.6	32.4
→ DPO → OT30K	10.0	6.7	20.0	36.4	51.4	46.3	11.9	40.8	26.8	11.5	43.0	27.7
→ OT30K (no Phase 1)	20.0	16.7	25.0	71.4	58.9	57.3	10.0	31.6	26.3	6.0	26.6	31.8
FineWeb-Edu-16k → SFT	0.0	0.0	2.5	1.4	8.5	2.4	0.3	2.2	30.3	3.0	27.6	7.1
→ OT30K	0.0	0.0	0.0	0.8	9.2	0.0	0.0	0.0	23.7	0.0	17.8	4.7
Nemotron-16k → SFT	0.0	0.0	2.5	1.4	6.4	1.5	0.2	0.2	25.9	3.3	36.5	7.1
→ OT30K	0.0	0.0	0.0	0.4	5.3	0.0	0.0	0.2	21.2	0.2	20.0	4.3
Comma0.1-16k → SFT	0.0	0.0	5.0	2.2	5.8	12.8	0.8	23.6	30.3	5.0	35.4	11.0
→ OT30K	0.0	0.0	0.0	1.6	14.0	1.8	0.0	0.0	28.8	0.0	20.0	6.0
SmolLM2-16k → SFT	0.0	3.3	2.5	6.6	31.9	28.7	3.3	37.4	25.8	6.5	45.0	17.4
→ OT30K	0.0	0.0	0.0	14.0	26.3	23.2	0.5	15.8	27.3	3.3	34.0	13.1
Qwen2.5 → SFT	0.0	0.0	7.5	19.6	39.7	48.2	3.8	44.6	22.7	7.5	44.4	21.6
→ OT30K	10.0	6.7	15.0	49.2	56.9	20.7	1.4	8.4	25.8	2.1	27.6	20.3
Qwen3 → SFT	10.0	0.0	12.5	23.0	69.2	64.6	3.8	55.8	25.8	11.7	53.2	30.0
→ OT30K	0.0	0.0	0.0	10.6	29.0	7.3	1.1	10.6	28.3	0.5	30.4	10.7

Table 11: Full held-out reasoning grid for MixtureVitae: four reasoning corpora applied from three starting checkpoints, with recipe, sample budget, and epoch count held fixed so that only the identity of the reasoning corpus changes. “—” marks a benchmark that was not evaluated, and that row’s average is taken over the remaining ten. Protocol B, the reasoning post-training protocol (Appendix Table 27). Single seed, so no error bars are available.

Reasoning corpus	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
<i>Starting from the post-YaRN base checkpoint</i>												
starting checkpoint	0.0	10.0	30.0	33.6	58.6	48.2	11.7	41.0	26.3	8.3	42.6	28.2
OpenThoughts3	20.0	16.7	25.0	70.2	59.1	57.3	10.0	31.6	26.3	6.0	26.6	31.7 (+3.5)
AceReason	3.3	13.3	30.0	69.2	61.8	52.4	9.8	35.6	26.3	10.4	24.6	30.6 (+2.4)
Nemotron-Cascade	3.3	6.7	27.5	70.0	60.0	49.4	11.4	35.8	26.8	10.2	29.0	30.0 (+1.8)
Superior-Reasoning	3.3	3.3	20.0	51.0	62.5	35.4	6.5	12.2	28.3	4.2	26.4	23.0 (−5.2)
<i>Starting from the Tulu3 SFT checkpoint</i>												
starting checkpoint	0.0	0.0	5.0	35.2	51.0	45.1	8.1	40.2	28.3	6.5	41.8	23.8
OpenThoughts3	6.7	10.0	20.0	70.6	50.7	58.5	10.0	29.2	36.4	4.6	25.6	29.3 (+5.5)
AceReason	3.3	20.0	25.0	69.4	54.4	48.2	10.8	33.6	27.3	9.9	25.6	29.8 (+6.0)
Nemotron-Cascade	3.3	13.3	27.5	66.8	52.6	50.6	—	32.8	34.8	7.3	30.6	32.0 (+8.2)
Superior-Reasoning	6.7	3.3	20.0	51.6	55.6	22.0	6.2	15.6	25.8	4.9	27.0	21.7 (−2.1)
<i>Starting from the Tulu3 SFT + DPO checkpoint</i>												
starting checkpoint	0.0	0.0	2.5	33.2	51.9	47.6	8.7	40.4	27.8	8.9	45.0	24.2
OpenThoughts3	10.0	6.7	20.0	36.4	50.7	44.5	11.9	40.8	26.8	11.5	43.0	27.5 (+3.3)
AceReason	6.7	16.7	27.5	70.8	55.4	48.2	11.4	32.6	23.7	9.8	25.0	29.8 (+5.6)
Nemotron-Cascade	6.7	10.0	25.0	68.2	53.5	50.0	13.3	34.8	28.8	9.7	30.4	30.0 (+5.8)
Superior-Reasoning	3.3	6.7	20.0	54.2	55.1	28.7	5.7	13.2	24.7	5.3	28.0	22.3 (−1.9)

948 **Model Architecture** Table 21 defines the two model scales used in our study. The columns are
949 defined as follows:

950 **Parameters (B) (Non-Emb + Emb)** The total model parameters in billions, separated into **Non-**
951 **Embedding** (Non-Emb) parameters (the core transformer blocks) and **Embedding** (Emb)
952 parameters (the token lookup tables). As noted in the caption, we used tied embedding
953 weights.

954 **Layers** The total number of transformer blocks stacked in the model.

Table 12: Per-benchmark results for the 100B-token MixtureVitae mixture-composition ablation on the primary reasoning suite. Removing the instruction-and-reasoning subset causes a broad collapse in math, code, and reasoning that Phase 1 Tulu3 SFT/DPO does not recover, whereas removing the curated web subset leaves performance comparable to, and on several benchmarks stronger than, the full mixture at both 4K and 16K context. Bold marks the column maximum.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
MV-noinstruct	0.0	0.0	0.0	0.4	2.6	0.6	0.0	14.2	23.9	0.0	24.8	6.1
→ SFT	0.0	0.0	2.5	1.8	4.2	7.3	1.0	15.8	20.7	3.4	29.9	7.9
→ SFT → DPO	0.0	0.0	2.5	2.0	3.8	7.0	1.4	15.2	25.1	4.0	32.0	8.5
MV-noinstruct-16k	0.0	0.0	2.5	1.6	4.2	1.5	0.0	2.0	24.6	4.7	32.9	6.7
→ SFT	0.0	0.0	0.0	2.2	4.8	3.4	0.4	3.4	25.6	2.1	35.2	7.0
→ SFT → DPO	0.0	0.0	2.5	2.6	5.2	4.9	0.6	4.8	23.7	2.2	36.0	7.5
MV-noweb	0.0	0.0	17.5	26.4	42.4	23.5	7.4	34.6	23.7	2.6	26.5	18.6
→ SFT	0.0	0.0	7.5	10.0	36.0	23.6	2.4	36.8	22.4	3.7	35.0	16.1
→ SFT → DPO	0.0	3.3	10.0	12.2	36.5	25.1	1.8	38.0	25.2	4.3	36.2	17.5
MV-noweb-16k	0.0	0.0	10.0	35.0	44.7	10.7	3.5	9.0	23.9	4.4	37.4	16.2
→ SFT	0.0	0.0	7.5	18.2	33.6	9.5	3.5	7.2	22.1	4.3	38.2	13.1
→ SFT → DPO	3.3	3.3	20.0	21.2	35.0	11.0	3.3	7.8	21.0	2.9	38.3	15.2
MV	3.3	0.0	0.0	12.2	42.1	0.6	0.0	0.2	24.4	3.2	21.2	9.8
→ SFT	0.0	0.0	5.0	6.6	38.4	22.3	3.3	36.8	20.7	2.5	34.9	15.5
→ SFT → DPO	0.0	0.0	5.0	8.0	39.2	22.6	4.3	36.4	24.4	2.9	35.1	16.2
MV-16k	0.0	0.0	17.5	35.2	40.6	2.5	3.7	7.2	22.4	7.2	40.4	16.1
→ SFT	0.0	0.0	0.0	3.2	36.5	11.6	0.6	6.4	23.9	4.4	39.5	11.5
→ SFT → DPO	0.0	0.0	0.0	3.4	36.4	0.9	0.8	8.2	24.6	4.0	41.6	10.9

Table 13: Reasoning-suite comparison between decontaminated and non-decontaminated 300B MixtureVitae checkpoints at 4K and 16K context, before and after Phase 1 Tulu3 SFT/DPO. Decontamination changes some base scores, but the post-training picture remains the same: the gains reported in the main paper persist after decontamination. Bold marks the column maximum.

Base	Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Not Decontaminated	MV	0.0	0.0	15.0	14.6	53.1	26.6	11.2	43.0	22.2	5.0	27.8	19.9
	→ SFT	6.7	0.0	2.5	16.4	50.6	32.5	11.2	42.6	25.6	3.5	45.0	21.5
	→ SFT → DPO	0.0	0.0	7.5	17.6	50.9	32.8	12.5	43.0	27.8	3.5	46.7	22.0
	MV-16k	3.3	6.7	20.0	36.6	61.8	30.0	16.8	41.8	22.6	11.0	47.9	27.1
	→ SFT	0.0	0.0	5.0	18.0	51.3	29.7	10.6	43.2	27.3	2.7	49.5	21.6
	→ SFT → DPO	0.0	0.0	0.0	18.0	51.5	30.0	11.0	41.8	25.2	3.8	50.7	21.1
Decontaminated	MV	0.0	10.0	7.5	17.4	53.5	11.7	12.7	18.6	23.9	4.1	25.6	16.8
	→ SFT	0.0	0.0	7.5	28.6	49.3	27.9	2.1	41.2	26.1	4.2	42.4	20.8
	→ SFT → DPO	0.0	0.0	15.0	27.4	50.5	27.9	1.8	40.8	24.6	3.1	42.9	21.3
	MV-16k	0.0	0.0	27.5	46.0	58.6	31.9	22.7	39.8	25.2	8.2	46.9	27.9
	→ SFT	3.3	0.0	12.5	31.0	51.9	33.2	15.5	40.6	23.4	5.2	47.1	24.0
	→ SFT → DPO	6.7	0.0	15.0	33.6	52.1	33.2	14.9	40.0	25.2	5.4	49.5	25.0

955 **Hidden** The hidden size (or embedding dimension, d_{model}) of the model.

956 **Heads** The number of attention heads in the multi-head attention mechanism.

957 **FFN Hidden** The inner dimension of the Feed-Forward Network (FFN) layer within each trans-
958 former block.

959 **Memory** The approximate VRAM required to store the model weights, in bfloat16.

960 **FLOPs** An approximation of the training compute cost using the **6N** rule: a standard estimate for
961 a transformer’s forward-and-backward pass, where **N** is the number of *non-embedding*
962 parameters [Kaplan et al., 2020].

963 Table 22 gives the hyperparameters for the reasoning post-training runs, and Tables 23 and 24 the
964 corresponding compute.

965 **Training Schedules** Table 20 defines the training hyperparameters for our two main experimental
966 runs (100B and 300B tokens). We use a single stage training with no post-training.

967 **Tokens** The total number of tokens in the training run.

968 **Global Batch Size (tokens)** The total number of tokens processed in a single training step (i.e., one
969 gradient update) across all GPUs.

Table 14: Reasoning-suite results for the smaller-scale comparison between MV (0.4B) and Baguettotron. Across 4K and 16K context, MV is substantially more receptive to Phase 1 post-training, especially on code and broader reasoning, while Baguettotron remains weak throughout. Bold marks the column maximum.

Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Baguettotron	0.0	0.0	0.0	0.2	2.3	0.0	0.0	0.0	23.1	0.2	24.3	4.5
→ SFT	0.0	0.0	2.5	2.0	7.7	0.0	0.0	0.0	24.4	1.3	26.9	5.9
→ SFT → DPO	0.0	0.0	0.0	2.8	8.3	0.0	0.0	0.0	26.3	1.1	30.2	6.2
Baguettotron 16k	0.0	0.0	0.0	2.0	5.2	0.6	0.0	0.0	21.6	1.8	34.2	5.9
→ SFT	0.0	0.0	2.5	2.4	8.1	0.0	0.0	0.0	24.4	1.1	30.0	6.2
→ SFT → DPO	0.0	0.0	2.5	3.2	8.7	0.0	0.0	0.0	24.9	1.5	31.8	6.6
MV (0.4B)	3.3	0.0	2.5	0.6	28.7	1.2	1.0	5.2	22.4	1.1	19.6	7.8
→ SFT	0.0	0.0	2.5	6.2	23.1	0.6	7.0	12.2	25.6	1.7	27.1	9.6
→ SFT → DPO	0.0	0.0	2.5	7.4	23.1	15.3	8.2	22.6	24.1	2.3	28.1	12.1
MV-16k (0.4B)	0.0	0.0	7.5	19.8	29.5	6.4	0.8	13.8	22.2	4.3	32.6	12.4
→ SFT	0.0	0.0	2.5	6.2	24.5	8.9	1.4	13.6	22.9	2.3	29.9	10.2
→ SFT → DPO	0.0	0.0	2.5	6.8	24.9	7.7	1.6	11.2	23.1	2.6	31.4	10.1

Table 15: Per-benchmark results for the 100B token MixtureVitae mixture composition ablation on the standard language understanding suite. Removing the instruction and reasoning subset creates a persistent language gap that Phase 1 does not close, whereas removing the curated web subset leaves language performance close to the full mixture across stages and both context lengths. Bold marks the column maximum.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
MV-noinstr	32.0	61.0	58.0	18.0	66.0	46.0	46.0	25.0	33.0	68.0	55.0	46.0
→SFT	32.0	61.0	57.0	19.0	66.0	46.0	47.0	25.0	33.0	68.0	55.0	46.0
→SFT →DPO	32.0	61.0	57.0	19.0	66.0	47.0	47.0	25.0	33.0	68.0	55.0	46.0
MV-noinstr-16k	33.0	60.0	63.0	19.0	69.0	46.0	47.0	25.0	33.0	67.0	55.0	47.0
→SFT	31.0	60.0	61.0	18.0	67.0	46.0	47.0	25.0	33.0	67.0	56.0	46.0
→SFT →DPO	31.0	60.0	61.0	18.0	67.0	47.0	47.0	25.0	32.0	67.0	56.0	46.0
MV-noweb	34.0	65.0	67.0	35.0	67.0	46.0	39.0	31.0	33.0	68.0	56.0	49.0
→SFT	33.0	63.0	65.0	35.0	69.0	47.0	41.0	33.0	32.0	67.0	55.0	49.0
→SFT →DPO	33.0	63.0	65.0	35.0	69.0	47.0	41.0	33.0	32.0	67.0	55.0	49.0
MV-noweb-16k	36.0	65.0	70.0	37.0	70.0	45.0	40.0	32.0	32.0	67.0	56.0	50.0
→SFT	33.0	63.0	66.0	34.0	69.0	46.0	41.0	33.0	33.0	67.0	56.0	49.0
→SFT →DPO	33.0	63.0	67.0	34.0	69.0	47.0	41.0	33.0	33.0	67.0	56.0	49.0
MV	34.0	65.0	71.0	31.0	67.0	48.0	44.0	32.0	34.0	68.0	58.0	50.0
→SFT	34.0	64.0	71.0	32.0	69.0	49.0	45.0	33.0	34.0	69.0	58.0	51.0
→SFT →DPO	34.0	64.0	71.0	32.0	70.0	49.0	45.0	33.0	33.0	69.0	58.0	51.0
MV-16k	35.0	65.0	68.0	30.0	67.0	47.0	45.0	33.0	33.0	68.0	57.0	50.0
→SFT	33.0	64.0	69.0	29.0	68.0	49.0	45.0	32.0	33.0	68.0	59.0	50.0
→SFT →DPO	33.0	64.0	69.0	30.0	68.0	49.0	45.0	32.0	33.0	68.0	59.0	50.0

970 **Iterations** The total number of training steps.

971 **Learning Rate** The peak learning rate used during training.

972 **Warmup** The number of initial *iterations* (steps) over which the learning rate linearly increases from
973 0 to its peak value.

974 **Cooldown (20%)** The number of final *iterations* (the last 20% of training) over which the learning
975 rate decays to zero.

976 **Compute resources** Compute resources spent on OpenThoughts3 30k experiments are shown in
977 table 21.

978 **Pre-training token budget** Tab. 25 shows the pre-training token budget for each model and its
979 ratio relative to MV.

980 D Evaluation setup details

981 Tables 26 and 27 define the reasoning-suite abbreviations together with the benchmark-specific
982 metrics, shot counts, decoding settings, and maximum generation lengths under each of the two

Table 16: Full language suite results for the main 1.7B comparison. Phase 1 Tulu3 SFT/DPO leaves language scores largely unchanged, with MixtureVitae matching FineWeb-Edu, trailing Nemotron, and remaining below the more heavily pretrained external baselines. Bold marks the column maximum.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Comma0.1-300BT	36.0	63.0	62.0	21.0	71.0	53.0	54.0	27.0	33.0	71.0	60.0	50.0
→SFT	37.0	65.0	64.0	20.0	74.0	54.0	57.0	27.0	33.0	72.0	61.0	51.0
→SFT →DPO	37.0	65.0	65.0	20.0	74.0	54.0	57.0	27.0	33.0	72.0	61.0	51.0
MV	40.0	71.0	74.0	51.0	71.0	54.0	48.0	42.0	34.0	70.0	58.0	56.0
→SFT	39.0	70.0	73.0	50.0	70.0	56.0	50.0	43.0	34.0	70.0	58.0	56.0
→SFT →DPO	39.0	70.0	73.0	50.0	71.0	56.0	50.0	43.0	34.0	70.0	58.0	56.0
MV-16k	42.0	71.0	73.0	49.0	67.0	53.0	49.0	41.0	32.0	70.0	58.0	55.0
→SFT	39.0	70.0	73.0	50.0	69.0	55.0	50.0	42.0	34.0	70.0	57.0	55.0
→SFT →DPO	39.0	71.0	73.0	50.0	69.0	55.0	50.0	42.0	34.0	70.0	57.0	55.0
Qwen2.5-1.5B	54.0	81.0	78.0	76.0	83.0	68.0	62.0	61.0	40.0	77.0	63.0	68.0
→SFT	51.0	79.0	70.0	77.0	82.0	68.0	60.0	60.0	40.0	77.0	63.0	66.0
→SFT →DPO	52.0	79.0	70.0	76.0	82.0	68.0	60.0	60.0	39.0	77.0	63.0	66.0
Qwen3-1.7B	54.0	82.0	83.0	76.0	82.0	67.0	63.0	63.0	38.0	76.0	64.0	68.0
→SFT	53.0	81.0	82.0	77.0	80.0	68.0	61.0	63.0	39.0	76.0	64.0	68.0
→SFT →DPO	53.0	81.0	83.0	77.0	80.0	68.0	61.0	63.0	39.0	76.0	64.0	68.0
SmolLM2-1.7B	54.0	80.0	75.0	60.0	82.0	73.0	67.0	50.0	44.0	79.0	67.0	66.0
→SFT	53.0	80.0	75.0	61.0	85.0	74.0	69.0	49.0	43.0	79.0	67.0	67.0
→SFT →DPO	54.0	80.0	75.0	61.0	85.0	74.0	69.0	49.0	43.0	79.0	67.0	67.0
FineWeb-Edu-300BT	48.0	77.0	65.0	21.0	82.0	67.0	56.0	26.0	42.0	77.0	62.0	57.0
→SFT	48.0	79.0	66.0	20.0	83.0	68.0	59.0	27.0	42.0	77.0	63.0	57.0
→SFT →DPO	48.0	78.0	66.0	20.0	83.0	68.0	59.0	27.0	42.0	77.0	63.0	57.0
Nemotron	52.0	81.0	77.0	49.0	85.0	71.0	58.0	46.0	42.0	79.0	65.0	64.0
→SFT	48.0	79.0	74.0	46.0	82.0	70.0	58.0	43.0	43.0	78.0	65.0	62.0
→SFT →DPO	48.0	79.0	74.0	46.0	82.0	71.0	59.0	43.0	43.0	78.0	65.0	63.0

Table 17: Language suite comparison between decontaminated and non-decontaminated 300B MixtureVitae checkpoints at 4K and 16K context, before and after Phase 1 Tulu3 SFT/DPO. In contrast to the reasoning suite, the language results are essentially unchanged by decontamination. Bold marks the column maximum.

Base	Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Not Decontaminated	MV	40.0	71.0	75.0	49.0	73.0	54.0	48.0	38.0	35.0	70.0	58.0	56.0
	→SFT	39.0	70.0	75.0	47.0	71.0	55.0	50.0	40.0	35.0	70.0	58.0	55.0
	→SFT →DPO	39.0	70.0	75.0	47.0	71.0	55.0	50.0	40.0	35.0	70.0	58.0	56.0
	MV-16k	39.0	71.0	75.0	50.0	72.0	54.0	49.0	38.0	36.0	71.0	58.0	56.0
	→SFT	39.0	71.0	74.0	48.0	72.0	55.0	50.0	38.0	35.0	70.0	59.0	55.0
	→SFT →DPO	39.0	71.0	74.0	48.0	72.0	55.0	50.0	38.0	35.0	71.0	59.0	56.0
Decontaminated	MV	40.0	71.0	74.0	51.0	71.0	54.0	48.0	42.0	34.0	70.0	58.0	56.0
	→SFT	39.0	70.0	73.0	50.0	70.0	56.0	50.0	43.0	34.0	70.0	58.0	56.0
	→SFT →DPO	39.0	70.0	73.0	50.0	71.0	56.0	50.0	43.0	34.0	70.0	58.0	56.0
	MV-16k	42.0	71.0	73.0	49.0	67.0	53.0	49.0	41.0	32.0	70.0	58.0	55.0
	→SFT	39.0	70.0	73.0	50.0	69.0	55.0	50.0	42.0	34.0	70.0	57.0	55.0
	→SFT →DPO	39.0	71.0	73.0	50.0	69.0	55.0	50.0	42.0	34.0	70.0	57.0	55.0

Table 18: Language suite results for the smaller scale comparison between MV (0.4B) and Baguet-totron. MV retains a small but consistent advantage across both context lengths and post-training stages, showing that its broader general-purpose edge at small scale is not limited to reasoning tasks. Bold marks the column maximum.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Baguet-totron	33.0	58.0	61.0	19.0	56.0	36.0	30.0	27.0	31.0	61.0	53.0	42.0
→SFT	30.0	57.0	61.0	21.0	61.0	36.0	33.0	26.0	32.0	61.0	55.0	43.0
→SFT →DPO	31.0	57.0	62.0	21.0	61.0	36.0	33.0	26.0	32.0	61.0	54.0	43.0
Baguet-totron 16k	32.0	58.0	57.0	21.0	60.0	36.0	34.0	26.0	31.0	61.0	55.0	43.0
→SFT	31.0	57.0	64.0	20.0	61.0	36.0	33.0	26.0	31.0	61.0	54.0	43.0
→SFT →DPO	31.0	57.0	64.0	19.0	61.0	36.0	33.0	26.0	32.0	61.0	54.0	43.0
MV (0.4B)	32.0	61.0	60.0	26.0	63.0	42.0	38.0	28.0	31.0	66.0	53.0	45.0
→SFT	29.0	59.0	50.0	25.0	63.0	42.0	40.0	29.0	30.0	65.0	53.0	44.0
→SFT →DPO	29.0	59.0	50.0	26.0	63.0	42.0	40.0	29.0	30.0	65.0	54.0	44.0
MV-16k (0.4B)	32.0	61.0	58.0	26.0	63.0	41.0	41.0	28.0	32.0	65.0	54.0	45.0
→SFT	28.0	59.0	53.0	25.0	64.0	42.0	40.0	29.0	31.0	65.0	55.0	45.0
→SFT →DPO	28.0	59.0	54.0	25.0	63.0	42.0	41.0	29.0	31.0	65.0	55.0	45.0

Table 19: Language suite results after Phase 2 OpenThoughts3 30K post-training for the models evaluated in the OT3 experiments. Reasoning post-training preserves general language understanding, and MV-16K performs equally well with the full pipeline or with direct OT3 post-training from the post-YaRN base. Bold marks the column maximum.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
MV-16k												
→SFT →DPO →OT30K	40.0	70.0	73.0	50.0	71.0	55.0	49.0	41.0	34.0	70.0	59.0	56.0
→OT30K (no Phase 1)	43.0	71.0	74.0	49.0	69.0	54.0	49.0	42.0	33.0	70.0	59.0	56.0
Comma-16k												
→SFT →OT30K	38.0	66.0	66.0	22.0	71.0	54.0	54.0	28.0	34.0	71.0	61.0	51.0
→SFT →DPO →OT30K	36.0	66.0	65.0	22.0	73.0	51.0	49.0	28.0	33.0	70.0	59.0	50.0
FineWeb-Edu-16k												
→SFT →OT30K	47.0	79.0	70.0	21.0	83.0	67.0	54.0	27.0	41.0	77.0	63.0	57.0
Nemotron-16k												
→SFT →OT30K	48.0	79.0	75.0	46.0	81.0	70.0	57.0	41.0	42.0	78.0	64.0	62.0
→SFT →DPO →OT30K	48.0	79.0	75.0	47.0	81.0	70.0	57.0	41.0	42.0	78.0	65.0	62.0
Qwen2.5-1.5B												
→SFT →OT30K	51.0	78.0	77.0	73.0	84.0	67.0	46.0	59.0	41.0	77.0	63.0	65.0
→SFT →DPO →OT30K	51.0	78.0	77.0	73.0	84.0	67.0	46.0	59.0	41.0	77.0	63.0	65.0
Qwen3-1.7B												
→SFT →OT30K	49.0	76.0	75.0	65.0	81.0	59.0	43.0	53.0	39.0	74.0	60.0	61.0
→SFT →DPO →OT30K	49.0	76.0	75.0	65.0	81.0	59.0	43.0	53.0	39.0	74.0	61.0	61.0
SmolLM2-1.7B-16k												
→SFT →OT30K	53.0	81.0	70.0	59.0	88.0	71.0	64.0	44.0	43.0	78.0	66.0	65.0
SmolLM2-1.7B-Instruct-16k												
→OT30K	51.0	79.0	76.0	62.0	83.0	71.0	59.0	47.0	44.0	78.0	66.0	65.0

Table 20: The training schedules used in our experiments.

Tokens	Global Batch Size (tokens)	Iterations	Learning Rate	Warmup	Cooldown (20%)
100B	4.12M	23,842	4×10^{-3}	8,334	4,768
300B	4.12M	72,661	4×10^{-3}	25,000	14,532

983 evaluation protocols, while Table 28 lists the standard language-understanding benchmarks and their
984 few-shot configurations under the default the LM Evaluation Harness setup.

Table 21: **open-sci-ref** [Nezhurina et al., 2026] model architecture and scales. We used tied embedding weights in all experiments.

Parameters (B) (Non-Emb + Emb)	Layers	Hidden	Heads	FFN Hidden	Memory	FLOPs
$0.35 + 0.05 = 0.40$	22	1024	16	3840	2.88 GB	2.4×10^9
$1.61 + 0.10 = 1.71$	24	2048	32	8192	9.884 GB	1.0×10^{10}

Table 22: Hyperparameters used for OT3 (OpenThoughts-3, 30k samples) SFT runs. All runs share the same recipe, and only the attention backend and chat template change per architecture.

Hyperparameter	Value
Dataset	open-thoughts/OpenThoughts-3 (30k samples)
Stage / fine-tuning type	SFT, full-parameter
Trainer / framework	LLaMA-Factory
Distributed strategy	DeepSpeed ZeRO-3
Precision	BF16
Liger Kernel	enabled
Attention backend	FA2 (Llama/Qwen) / SDPA (opensci-arch)
Sequence length (cutoff_len)	16,384
Per-device batch size	1
Gradient accumulation steps	4
World size	32 H100 (8 nodes $\times$ 4)
Effective global batch size	128
Optimizer	AdamW (LLaMA-Factory default)
Learning rate	4×10^{-5}
LR schedule	Cosine
Warmup ratio	0.1
Epochs	5
Group by length	true
Seed	42

Table 23: Single-run GPU-hours for OT3 (30k) SFT. All runs: 32 H100 (8 nodes $\times$ 4 GPUs). Difference between groups is the attention backend: FA2 (Llama/Qwen) is $\sim 2\times$ faster than SDPA (opensci-arch).

Base model	Attention	GPU-hours	Training
MixtureVitae-1.7B	SDPA	106	OT3 30K SFT
MixtureVitae-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
MixtureVitae-1.7B + DPO-Tulu3	SDPA	105	OT3 30K SFT
Comma0.1-1.7B + SFT-Tulu3	SDPA	105	OT3 30K SFT
Comma0.1-1.7B + DPO-Tulu3	SDPA	158	OT3 30K SFT
FineWeb-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
Nemotron-1.7B + SFT-Tulu3	SDPA	106	OT3 30K SFT
Nemotron-1.7B + DPO-Tulu3	SDPA	106	OT3 30K SFT
Qwen3-1.7B-Base	FA2	60	OT3 30K SFT
Qwen3-1.7B-Base + SFT-Tulu3	FA2	57	OT3 30K SFT
Qwen3-1.7B-Base + DPO-Tulu3	FA2	57	OT3 30K SFT
Qwen2.5-1.5B	FA2	53	OT3 30K SFT
Qwen2.5-1.5B + SFT-Tulu3	FA2	53	OT3 30K SFT
Qwen2.5-1.5B + DPO-Tulu3	FA2	52	OT3 30K SFT
SmolLM2-1.7B	FA2	48	OT3 30K SFT
SmolLM2-1.7B-Instruct	FA2	48	OT3 30K SFT
SmolLM2-1.7B + SFT-Tulu3	FA2	44	OT3 30K SFT
SmolLM2-1.7B + DPO-Tulu3	FA2	47	OT3 30K SFT
Total (18 runs)		1417	

Table 24: Single-run GPU-hours for long-context extension SFT. All runs: 8 GH200 (2 nodes $\times$ 4 GPUs), 1 epoch ($\approx 16k$ tokens/seq, YaRN RoPE scaling), SDPA attention.

Base model	Attention	GPU-hours	Training
MixtureVitae-1.7B	SDPA	12	Long-ctx SFT (16k)
Comma0.1-1.7B	SDPA	12	Long-ctx SFT (16k)
DCLM-1.7B	SDPA	12	Long-ctx SFT (16k)
FineWeb-Edu-1.7B	SDPA	12	Long-ctx SFT (16k)
Nemotron-HQ-1.7B	SDPA	12	Long-ctx SFT (16k)

Table 25: Pre-training token budget for each model and its ratio relative to MV (300B tokens). External open-weights references use substantially more pretraining compute than MV.

	{MV,Comma0.1,FW-Edu,Nemotron}	SmolLM2	Qwen2.5-1.5B	Qwen3-1.7B
# Tokens (T)	0.3	11	18	36
Budget Ratio	1.0	36.7	60.0	120.0

D.1 Evaluation protocols

Two evaluation protocols are used, and no value produced under one is ever compared against a value produced under the other. Protocol A (Table 26) decodes greedily on every benchmark under one fixed generation budget, which is what makes thirteen model families and three decoding seeds comparable, and it covers Phase 1, the 300B factorial, and the mixture ablations. Protocol B (Table 27) uses sampled decoding and longer generation budgets on several benchmarks, because models that have received reasoning post-training emit long chains of thought that a short greedy budget truncates, and it covers every OpenThoughts3 experiment. The same checkpoint scores differently under the two, which is why every result in Section 5 is reported as a paired change against a starting checkpoint re-evaluated under Protocol B.

Listed token lengths are upper bounds and are reduced accordingly for models with a shorter context window. Chat benchmarks are run through `evalchemy` and GSM8K uses the standard the LM Evaluation Harness `gsm8k` setup.

Table 26: **Protocol A, the unified evaluation protocol.** Used for Phase 1 across the model zoo, for the 300B front-loading factorial, and for the mixture ablations. All results under this protocol are reported as mean and sample standard deviation over decoding seeds 42, 1234, and 2025.

Abbr.	Task	Ref.	Metric	#Shots	Decoding	Max new toks
A24	AIME24	[MAA, 2024]	Accuracy	0	Greedy	32,768
A25	AIME25	[MAA, 2025]	Accuracy	0	Greedy	32,768
AMC	AMC23	[MAA, 2023]	Accuracy	0	Greedy	32,768
M500	MATH500	[Hendrycks et al., 2021]	Accuracy	0	Greedy	32,768
GSM8K	GSM8K	[Cobbe et al., 2021]	Exact match	5	Greedy	2,048
HE	HumanEval	[Chen et al., 2021]	<code>python_pass@1</code>	0	Greedy	2,048
LCB	LiveCodeBench (v2)	[Jain et al., 2025]	Pass@1	0	Greedy	32,768
MBPP	MBPP	[Austin et al., 2021]	Functional Pass@1	3	Greedy	2,048
GPQA	GPQA-Diamond	[Rein et al., 2024]	Accuracy	0	Greedy	32,768
JEE	JEEBench	[Arora et al., 2023]	Accuracy (JEE score)	0	Greedy	32,768
IFE	IFEval	[Zhou et al., 2023b]	Instruction-level accuracy	0	Greedy	2,048

Table 27: **Protocol B, the reasoning post-training protocol.** Used for every OpenThoughts3-30K experiment, namely the endpoints of Table 10, the pre-training ablation of Table 3, and the held-out reasoning grid of Tables 4 and 11. All benchmarks use a single generation per problem (`n_repeat=1`) and a single seed. “Decoding” is the effective decoding after `lm_eval`’s `normalize_gen_kwargs: do_sample=False` is normalised to greedy even when a nonzero temperature is set in the source, while `do_sample=True` samples at the listed temperature.

Abbr.	Task	Ref.	Metric	#Shots	Decoding	Max new toks
A24	AIME24	[MAA, 2024]	Accuracy	0	Greedy	32,768
A25	AIME25	[MAA, 2025]	Accuracy	0	Sampling ($T=0.6, p=0.95$)	32,768
AMC	AMC23	[MAA, 2023]	Accuracy	0	Greedy	32,768
M500	MATH500	[Hendrycks et al., 2021]	Accuracy	0	Sampling ($T=0.7$)	32,768
GSM8K	GSM8K	[Cobbe et al., 2021]	Exact match	5	Greedy	256
HE	HumanEval	[Chen et al., 2021]	Pass@1	0	Sampling ($T=0.6$)	16,000
LCB	LiveCodeBench (v5)	[Jain et al., 2025]	Pass@1	0	Greedy	32,768
MBPP	MBPP	[Austin et al., 2021]	Pass@1	0	Greedy	512
GPQA	GPQA-Diamond	[Rein et al., 2024]	Accuracy	0	Sampling ($T=0.7$)	32,768
JEE	JEEBench	[Arora et al., 2023]	Accuracy (JEE score)	0	Greedy	32,768
IFE	IFEval	[Zhou et al., 2023b]	Prompt-level accuracy	0	Greedy	512

E Decontamination

Front-loading mathematics, code, and reasoning traces into pre-training is the regime in which benchmark leakage is most likely, so the decontamination control matters for every claim in this paper.

Procedure. The MixtureVitae pre-training corpus and all shared post-training data are decontaminated against every benchmark reported here using the 13-gram exact-match protocol of Nguyen et al. [2025]. Decontamination is applied separately to the pre-training corpus, to the Tulu3 SFT mixture, to the Tulu3 preference data, and to the 30K OpenThoughts3 subset used for reasoning

Table 28: Standard language understanding evaluation benchmarks used in our result tables. All tasks follow the default the LM Evaluation Harness [Gao et al., 2021] configurations with accuracy as the primary metric.

Abbr.	Task	Ref.	#Shots
MMLU	MMLU	[Hendrycks et al., 2020]	5
HellaS	HellaSwag	[Zellers et al., 2019]	10
CSQA	CommonSenseQA	[Talmor et al., 2019]	10
ARC-C	ARC-Challenge	[Clark et al., 2018]	10
ARC-E	ARC-Easy	[Clark et al., 2018]	10
PIQA	PIQA	[Bisk et al., 2020]	10
BoolQ	BoolQ	[Clark et al., 2019]	10
WinoG	Winogrande	[Sakaguchi et al., 2021]	0
OBQA	OpenBookQA	[Mihaylov et al., 2018]	0
COPA	COPA	[Roemmele et al., 2011]	0
LAMBADA	LAMBADA	[Paperno et al., 2016]	0

post-training. Because the same decontaminated post-training data is used for every model, no model in the comparison receives an advantage from post-training-side leakage.

An asymmetry that favors the references. The non-permissive substrates and the external open-weights references are used as released and inherit whatever contamination their own pre-training carries. We could not decontaminate them without retraining them, so the comparison is asymmetric in their favor, and any residual leakage in those corpora inflates their scores rather than ours.

Effect on the reported numbers. Tables 13 and 17 compare decontaminated against non-decontaminated 300B checkpoints. The largest movement is at the 4K base stage, where decontamination costs about three points of reasoning suite average, while the 4K post-trained scores move by well under a point and the 16K post-trained scores are higher after decontamination, by 2.4 points after SFT and 3.9 after DPO. Language understanding is unchanged within rounding. Decontamination therefore removes a measurable but small part of the base-stage score and does not account for the post-training differences this paper reports.